

RETAILSYNC

Centralized Super Shop Warehouse Management System

"Your retail inventory, synchronized and secure."

Comprehensive Project Proposal & Operational Study
An Automated Retail Warehouse Management & Inventory Optimization System
with Real-Time POS Sync, FEFO/FIFO Batch Tracking, and Automated Reordering Decision Support

PROJECT PROPOSAL

PART 1: PROJECT PLANNING AND DEFINITION

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PART A

THE PROBLEM & REAL-WORLD CONTEXT

1. Project Overview

RetailSync is a comprehensive Warehouse Management System (WMS) engineered specifically for mid-to-large tier super shops and grocery retail chains in Bangladesh. By establishing a real-time, deterministic data bridge between the back-end warehouse operations (receiving, batching, quarantine, putaway, auditing) and the front-end Point-of-Sale (POS) registers across retail floors, RetailSync eliminates the operational blindness, phantom inventory, and perishable waste that plague modern grocery retail.

Project Vision Statement

"To empower retail managers with deterministic, real-time inventory data, reducing perishable waste by up to 30%, eliminating undetected stockouts, and automating supplier reordering processes through an agile, synchronized, and mathematically optimized warehouse ecosystem."

Product Philosophy: The foundational philosophy of RetailSync is that warehouse inventory data must never remain a passive static record or live on isolated paper clipboards. Instead, every physical movement of goods—from dock receiving to spatial putaway, retail shelf replenishment, and cashier barcode scan—must instantly trigger active operational intelligence, autonomous stockout warnings, automated purchase requisition drafts, and strict shelf-life priority picking.

Key Terminology: Throughout this project proposal, the following standard terminology is adopted: 'Super Shop' refers to modern self-service grocery and department retail chains (e.g., Shwapno, Agora, Meena Bazar, Unimart, Daily Shopping); 'Central Distribution Center (CDC)' designates the primary warehouse hub supplying retail branches; 'SKU' represents a distinct Stock Keeping Unit; 'GRN' is the Goods Receipt Note validating supplier deliveries; 'FEFO' denotes First-Expired, First-Out batch picking logic; 'FIFO' denotes First-In, First-Out; 'EOQ' represents Economic Order Quantity; and 'ROP' indicates Dynamic Reorder Point threshold.

2. Real-World Operational Study & Background

2.1 The Bangladeshi Retail Challenge

The urban retail landscape across Bangladesh—from Dhaka and Chittagong to major regional hubs—has shifted rapidly from traditional open-air wet markets ('Kacha Bazar') and fragmented corner grocers ('Mudir Doka') to organized, structured super shops. However, the operational backend of many of these stores still relies on localized XAMPP/MySQL setups running isolated, standalone POS software that does not communicate with the warehouse ledger.

A standard mid-sized super shop in Bangladesh routinely grapples with severe operational complexities:

- **Supplier Volume:** 50+ independent suppliers delivering goods daily across varying schedules with unpredictable lead times.
- **Catalog Scale:** Over 5,000 unique SKUs (Stock Keeping Units) spanning dry food, household goods, chilled dairy, and fresh produce.

- **Perishable Sensitivity:** Highly perishable goods (pasteurized milk, yoghurt, meat, bakery items) that mandate strict First-Expired, First-Out (FEFO) and First-In, First-Out (FIFO) handling.
- **Checkout Concurrency:** High foot traffic during peak hours (weekends and religious festival seasons like Ramadan), requiring cashier checkout speeds of less than 3 seconds per item.

2.2 The Financial Impact & Three Major Financial Leaks

Without an integrated, synchronized Warehouse Management System, super shops face three major financial leaks that heavily erode profit margins:

- **1. Phantom Inventory:** The POS system indicates that there are 10 bottles of cooking oil in stock, but they were stolen, broken, or misplaced in the warehouse. Customers looking for them in-store or placing online delivery orders are disappointed, leading to direct lost sales, cancelled orders, and severe reputational damage.
- **2. Expiry Shrinkage:** Because warehouse staff manually stack shelves without systematic batch routing, older batches of dairy or packaged foods are pushed to the back of racks. They expire unnoticed, resulting in a 100% loss of capital on those expired items and risking severe food safety regulatory penalties.
- **3. The Procurement Bottleneck:** Store managers spend 3 to 4 hours every night manually walking down the aisles with a clipboard to visually inspect shelves and determine what needs to be ordered for the next day. This manual process is heavily error-prone, subjective, and creates massive operational bottlenecks.

2.3 Market Context: Empirical Telemetry & Bangladesh Market Metrics

To ground the design of RetailSync in empirical reality, official statistical indicators from the Bangladesh Bureau of Statistics (BBS), supermarket industry reports, and retail logistics telemetry were synthesized:

Market & Logistics Indicator	Empirical Baseline Metric in Bangladesh Context
Urban Population in Bangladesh	Approx. 65.2 million people (39.7% of total population, Census 2022 / BBS). Rapid urbanization driving demand for centralized retail.
Modern Trade Grocery Share	Currently ~2.5% to 3.2% of national grocery trade; projected to reach 10-12% by 2030 with a 16.5% CAGR in major metropolitan zones (Dhaka, Chittagong, Sylhet).
FMCG Retail Wastage & Spoilage	15% to 22% of perishable goods (dairy, fresh produce, meat, chilled foods) wasted annually across retail supply chains due to lack of FEFO batch tracking.
Retail Stockout Revenue Loss	Estimated 7.5% to 11.2% in unrealized revenue caused by sudden out-of-stock events on high-velocity FMCG items and delayed procurement reorders.
Inventory Shrinkage Rate	Average 1.8% to 2.4% of total inventory value lost to unrecorded shelf damages, theft, phantom inventory counts, and untracked supplier delivery discrepancies.
Central Warehouse Lead Time	Average order-to-shelf replenishment cycle takes 48 to 72 hours from central distribution centers to retail outlets due to manual picking and sorting bottlenecks.

Market & Logistics Indicator	Empirical Baseline Metric in Bangladesh Context
Barcode / Scanning Penetration	Over 95% of packaged FMCG items carry standard GS1/EAN barcodes, but warehouse putaway and picking remains largely manual paper-manifest driven.

These empirical indicators underscore three vital conclusions: First, the rapid 16.5% CAGR in organized supermarket retail demands automated, scalable backend systems. Second, the 15% to 22% annual spoilage rate on perishables represents an unsustainable drain that can be prevented through algorithmic batch expiration tracking. Third, existing software solutions in Bangladesh are overwhelmingly front-end billing POS tools that completely ignore back-end warehouse coordination.

Market Sizing (TAM/SAM/SOM): The Total Addressable Market (TAM) encompasses the 65+ million urban population served by retail grocery chains across Bangladesh. The Serviceable Addressable Market (SAM) comprises approximately 450+ central distribution warehouses and regional hubs operated by supermarket chains and FMCG distributors (e.g., Shwapno, Agora, Meena Bazar, Unimart, Daily Shopping, Chaldal). The Serviceable Obtainable Market (SOM) targets 10 to 15 pilot warehouse facilities during initial capstone roll-out.

3. Problem Statement & Feature Mapping

3.1 Core Operational Problems & Solution Mapping

The operational study identifies five foundational problems in Bangladeshi retail logistics, directly mapped to RetailSync solutions:

ID	Real-World Problem	Operational Impact	RetailSync Solution
P-01	Data Isolation	Sales data lives in the POS; warehouse data lives on clipboards.	Centralized SQL Database serving both POS and Warehouse APIs.
P-02	Undetected Stockouts	High-demand items run out on shelves before restock is ordered.	Automated Reorder Point (ROP) triggers sending alerts to managers.
P-03	Perishable Waste	Lack of batch tracking leads to expired goods.	Strict FIFO/FEFO logic and expiry-date dashboard highlights.
P-04	Supplier Delays	Manual PO generation is slow and error-prone.	Automated PO generation based on low-stock metrics.
P-05	Internal Shrinkage	Untraceable manual stock adjustments.	Role-Based Access Control (RBAC) with immutable audit logging.

3.2 Detailed Engineering Problem Taxonomy (P-01 to P-09)

To ensure comprehensive architectural coverage and academic rigor, these challenges are further decomposed into nine permanent engineering problem statements (P-01 to P-09), establishing full traceability across requirements, algorithms, and test cases:

ID	Problem Domain	Operational Reality & Pain Points	RetailSync Architectural Response
P-01	Disjointed Inbound Receiving and Error-Prone Manual GRN Verification	Inbound supplier deliveries are checked against paper purchase orders, leading to undetected quantity discrepancies, damaged packaging acceptance, and slow Goods Receipt Note (GRN) generation.	Digital Inbound Verification with handheld barcode scanning, real-time PO reconciliation, and automated digital GRN generation with variance capture.
P-02	Warehouse Opacity and Blind Spatial Putaway (Bin Search Latency)	Floor workers place received pallets in random, unindexed warehouse locations without systematic bin mapping, causing severe search latency and misplaced stock during picking.	Directed Spatial Putaway Engine with real-time Zone-Aisle-Rack-Shelf-Bin spatial mapping and capacity-aware putaway suggestions.
P-03	Perishable Spoilage and Financial Loss from Non-FEFO Picking	Warehouse operators pick goods on a naive LIFO or convenient nearest-bin basis rather than earliest expiry, resulting in billions of BDT in spoiled dairy, juices, and packaged food.	Strict FEFO (First-Expired, First-Out) batch allocation algorithm that dynamically directs pickers to the batch with the nearest expiration date.
P-04	Inventory Shrinkage, Unreconciled Discrepancies, and Phantom Stock	Physical stock levels consistently diverge from spreadsheet records due to unrecorded breakages, internal theft, and paper tally omissions, leading to 'phantom inventory'.	Continuous Cycle Counting Module, immutable audit ledgers, and machine-learning-assisted anomaly/shrinkage detection.
P-05	Inaccurate Stock Replenishment: The Overstocking vs. Understocking Trap	Procurement officers rely on intuitive guesswork, causing capital lockup in slow-moving items and disastrous stockouts during peak promotional weekends and holidays.	Algorithmic Replenishment Engine integrating Economic Order Quantity (EOQ), Greasley's Statistical Safety Stock, and Dynamic Reorder Point (ROP).

ID	Problem Domain	Operational Reality & Pain Points	RetailSync Architectural Response
P-06	Outbound Picking Inefficiencies and Order Assembly Bottlenecks	Outbound orders for multi-branch retail outlets are picked piece-by-piece with redundant travel paths across warehouse aisles, causing dock congestion and late store delivery.	Optimized Wave and Batch Picking algorithms generating shortest-path picker routing and multi-store sortation staging.
P-07	Supplier Performance and Lead-Time Volatility Opacity	Retailers have zero telemetry on supplier delivery timeliness, lead-time variance, or product rejection rates, crippling negotiation leverage and safety stock precision.	Supplier SLA Performance Scorecards tracking fulfillment accuracy, lead-time standard deviation, and historical damage rates.
P-08	Information Silos Between Central Warehouses and Branch Super Shops	Retail store managers cannot see real-time warehouse stock positions, leading to duplicate emergency store requisitions and uneven stock distribution across branches.	Unified Multi-Store Requisition Portal with real-time central stock visibility, store stock reservation, and in-transit dispatch tracking.
P-09	Regulatory Non-Compliance and Lack of Immutable Audit Ledgers	Failure to document batch origins and cold-chain compliance violates the Bangladesh Food Safety Act 2013 and BSTI standards, risking heavy penalties and brand damage.	Complete Batch-to-Store Traceability, digital chain-of-custody logging, and automated compliance reporting.

4. Existing Solution Gap & Competitor Comparison

To establish the distinct necessity of RetailSync, existing software tools and operational approaches in Bangladesh were analyzed. Currently, retail enterprises resort to four unsatisfactory compromises:

- **Manual Spreadsheets & Paper Clipboards:** Fragile, prone to data corruption, zero concurrency control, lacking spatial bin indexing or automated reorder logic.
- **Standalone POS Software (e.g., PrismPOS, Tally):** Optimized exclusively for customer retail billing and cashier cash-drawers; completely blind to warehouse aisle/rack topology, batch expiry tracking, or supplier lead-time variance.
- **Legacy Tier-1 ERPs (e.g., SAP, Oracle):** Cost-prohibitive (millions of BDT in licensing and annual maintenance), multi-year deployment cycles, rigid workflows unsuited for local super shop operational realities.
- **Mid-Market Accounting Tools (e.g., PrismERP):** Primarily financial accounting packages with basic stock count fields; lacking mobile barcode scanning, real-time spatial putaway, or FEFO pick routing.

4.1 Competitor Capability Comparison Matrix

The comprehensive capability matrix below contrasts existing solutions against the proposed RetailSync system across ten vital operational dimensions:

Capability / Operational Dimension	Manual Sheets / Paper	Standalone POS (e.g. PrismPOS, Tally)	Legacy Tier-1 ERP (e.g. SAP / Oracle)	Proposed RetailSync
Real-Time Barcode Inbound & GRN	No (Manual logbooks)	Limited (POS-centric)	Yes (High complexity)	Yes (Intuitive, fast mobile UI)
Dynamic Spatial Bin & Rack Mapping	No (Worker memory)	No (Only SKU counts)	Yes (Rigid setup)	Yes (Visual 2D/3D zone bin map)
Enforced FEFO Expiry Allocation	No (Frequent spoilage)	Rarely (Static FIFO only)	Yes (Custom configuration)	Yes (Native FEFO priority queue)
Greasley Statistical Safety Stock	No (Arbitrary min-max)	No (Static thresholds)	Partial (Requires manual tuning)	Yes (Dynamic statistical model)
Automated EOQ & Dynamic ROP	No (Human guesswork)	No (Basic reorder alert)	Yes (Complex add-on module)	Yes (Native DSS engine)
Wave & Batch Picking Routing	No (Random walking)	No (Single order billing)	Yes (Enterprise tier)	Yes (Optimized pick-path wave)
Multi-Branch Store Requisitions	No (Phone calls / email)	Limited (Separate databases)	Yes (Centralized)	Yes (Unified branch requisition portal)
Cycle Counting & Discrepancy Audit	No (Annual painful audit)	Partial (End-of-day tally)	Yes (Formal audit)	Yes (Continuous cycle counting)
Total Cost & Implementation Agility	Low cost / high waste	Low / Siloed functionality	Extremely expensive (Crores BDT)	Cost-effective, modern, modular

The comparison illustrates that no existing solution in the Bangladeshi market provides an integrated, cost-effective, and mobile-accessible platform combining real-time POS synchronization, spatial putaway, algorithmic FEFO batch picking, and mathematical replenishment optimization. Part B details the architecture and operational models that resolve this gap.

PART B

PROPOSED SOLUTION & ARCHITECTURE

5. Product Positioning and Differentiation

RetailSync moves beyond basic inventory bookkeeping. Its superiority stems from uniting spatial warehouse intelligence, sub-second POS inventory deduction, strict perishable expiry lifecycle enforcement, and mathematical decision support into a unified, user-friendly cyber-physical architecture tailored for supermarket supply chains.

5.1 Why RetailSync Is Different

Operational Dimension	Typical Retail Warehouse Software	RetailSync Value Proposition
Putaway Strategy	Static bin allocation; items dumped wherever space is found.	Directed spatial putaway based on product category, velocity (ABC classification), and bin capacity limits.
Perishable Handling	Naive FIFO (First-In, First-Out) or unmanaged picking causing expiry loss.	Strict FEFO (First-Expired, First-Out) batch routing, proactive expiry alert windows, and automated quarantine locks.
Inventory Replenishment	Manual reordering based on panic when shelves empty.	Mathematical DSS calculating dynamic EOQ, Greasley Safety Stock, and ROP factoring in supplier lead-time variance.
Outbound Assembly	Single-order manual picking with excessive walking time.	Consolidated wave/batch picking with intelligent pick-path ordering to minimize travel distance across aisles.
Store-Warehouse Sync	Store managers call or email central DC with unverified orders.	Dedicated Store Requisition Portal with live stock visibility, automated allocation, and dispatch delivery notes.
Operational Usability	Bloated, complex enterprise interfaces requiring months of training.	Clean, responsive mobile/tablet web UI with rapid barcode gun scanning and intuitive visual dashboards.

5.2 Formal Product Positioning Statement

Formal Product Positioning Statement

"For retail super shop chains and central grocery distribution hubs that struggle with warehouse opacity, perishable spoilage, and stockout bottlenecks, RetailSync is a centralized, mobile-ready Warehouse Management & Inventory Optimization System. Unlike generic spreadsheets, standalone POS software, or rigid multi-crore ERPs, RetailSync delivers real-time spatial bin putaway, automated FEFO expiry allocation, sub-2.0s POS inventory deduction, and mathematical replenishment decision support (EOQ, Greasley Safety Stock, ROP)

within a lightweight, highly usable, and cost-effective modern web platform."

5.3 Phased Market Entry & Deployment Strategy

To guarantee operational feasibility and de-risk deployment, RetailSync adopts a structured 3-stage market entry strategy: Stage 1 (Capstone MVP & Pilot) deploys the core system in a central distribution warehouse serving 3 retail branch outlets, validating barcode scanning, directed putaway, and FEFO picking on high-velocity FMCG and dairy items. Stage 2 expands functionality to include multi-temperature regional distribution centers and advanced supplier scorecards. Stage 3 transitions into a commercial enterprise SaaS/On-Premise solution supporting nationwide supermarket chains.

6. Proposed Solution Architecture, Workflows & Operational Model

RetailSync operates as a multi-tier cyber-physical software platform that captures physical stock movements via handheld barcode readers, enforces ACID transactional ledger integrity in a relational database, executes optimization algorithms in real time, and renders actionable telemetry on responsive web dashboards.

6.1 Step-by-Step Physical to Digital Workflow (4 Core Steps)

To understand RetailSync, the physical operational realities of a super shop are mapped directly to digital system workflows:

- Step 1: Supplier Delivery & Receiving (GRN):** Physical Reality: A supplier delivery truck arrives with 50 cartons of milk. The inventory clerk physically inspects and counts them.
 Digital Action: The clerk opens RetailSync on a mobile terminal, selects the active Purchase Order (PO), and generates a Goods Receipt Note (GRN). The clerk inputs the specific Expiry Date of this batch. The system updates 'Available' stock in real-time.
- Step 2: Quarantine & Damage Handling:** Physical Reality: During unloading, 2 cartons are found crushed or leaking.
 Digital Action: The clerk flags these 2 cartons as 'Damaged/Quarantine' in RetailSync. They are logged for supplier credit return and are strictly NOT added to the sellable POS inventory.
- Step 3: The Retail Floor & POS Sync:** Physical Reality: A customer purchases 3 cartons of milk at the retail checkout counter.
 Digital Action: The cashier scans the barcode. The POS sends an API request to RetailSync. The system uses FIFO/FEFO logic to deduct 3 units from the OLDEST active batch of milk in the database. Latency is guaranteed under 2.0 seconds.
- Step 4: Automated Auditing & Reordering:** Physical Reality: The total milk stock on shelves and in the backroom drops to 10 cartons (the minimum safe limit).
 Digital Action: RetailSync detects the Reorder Point (ROP) breach. It automatically drafts a Purchase Order for the specific dairy supplier and dispatches a dashboard notification to the Store Manager for one-click approval.

6.2 End-to-End 7-Stage Warehouse Material Flow

Expanding upon the 4 core steps, the comprehensive material and information flow across central distribution warehouses follows seven tightly coordinated stages:

- **1. Inbound Receiving & Digital GRN:** Inbound supplier deliveries are validated against digital PO line items via handheld barcode interrogation; damaged items logged; internal batch labels printed; digital GRN issued.
- **2. Directed Spatial Putaway:** SKU velocity (ABC classification) and temperature constraints guide operators to the optimal Zone-Aisle-Rack-Shelf-Bin; shelf barcode scan commits placement.
- **3. Relational Ledger Commit:** Atomic database transaction increments physical on-hand stock and registers the batch into the real-time FEFO priority queue.
- **4. Outbound Requisition & Wave Allocation:** Branch stores submit daily requisitions; RetailSync aggregates orders into picking waves, allocating stock from earliest-expiring available batches.
- **5. Shortest-Path Wave Picking:** Warehouse operators receive digital pick lists sorted by shortest travel path; pickers scan source bins and items to verify batch accuracy.
- **6. Staging & Store Dispatch:** Picked items are sorted into store-specific rolling cages; an official Delivery Chalan / Dispatch Note is printed for delivery transit.
- **7. Store Receipt Reconciliation:** Store receiving staff scan incoming totes against the dispatch manifest, confirming receipt and closing the requisition cycle.

6.3 Technical Architecture & Stack Recommendation

To ensure high performance, bulletproof concurrency, and data integrity, RetailSync relies on an industry-standard, robust technology stack:

Component	Recommended Technology	Justification for Capstone
Database Layer	MySQL / PostgreSQL	Relational architecture is mandatory for retail to ensure ACID compliance (Atomicity, Consistency, Isolation, Durability) for financial transactions.
Backend API	Python (Django/FastAPI) or Java	Excellent handling of concurrent requests, complex rule engines, and automated task scheduling (e.g., nightly expiry checks).
Frontend Dashboard	React / Next.js	Provides a highly responsive, single-page application (SPA) experience for managers, eliminating page-reload delays.
Local Testing Env.	XAMPP / Beekeeper Studio	Standard, accessible environment for developing and visualizing the relational schema during the capstone phase.

6.4 Conceptual 4-Tier System Architecture

The complete RetailSync software architecture is structured across four clean, decoupled layers: 1. Physical Edge & Data Capture Layer: Handheld Bluetooth/USB barcode scanners and smartphone cameras capturing 1D/2D GS1 barcodes on pallets, cartons, and shelf bins. 2. API Gateway & Ingestion Service Layer: Stateless RESTful services (Node.js/Express or Python FastAPI) managing scan validation, payload parsing, and JWT authentication. 3. Relational Ledger & Analytics Data Layer: PostgreSQL 16 / MySQL database enforcing strict ACID compliance, foreign key referential integrity, B-Tree indexes, and append-only audit ledgers, coupled with Redis for high-speed

caching. 4. Presentation & Visualization Layer: Responsive Next.js / React web portal providing live floor telemetry, 2D interactive spatial bin maps, mobile scanner views, and executive KPI analytics.

6.5 Production-Ready Relational Database Schema Blueprint (DDL & ERD Design)

To satisfy the rigorous academic data modeling standards of SE-231 (Software System Analysis & Design), RetailSync enforces strict Third Normal Form (3NF) relational integrity across eight production-grade core entities. The schema implements declarative foreign key cascading controls, domain value check constraints, timestamped audit triggers, and strategic B-Tree indexing tailored for sub-millisecond barcode scan queries and atomic POS stock deduction:

Table Name	Primary Purpose & Operational Entity	Primary Key & Unique Keys	Key Foreign Key Constraints	Index Optimization Strategy
suppliers	Supplier directory, contact terms, lead-time variance metrics for Greasley safety stock.	supplier_id (PK)	None (Root entity)	B-Tree on company_name, email
products	Master SKU catalog, category classification, storage temperature, unit & holding costs.	product_id (PK), sku_code (UQ), barcode_ean (UQ)	supplier_id -> suppliers(supplier_id)	B-Tree on sku_code, barcode_ean, category
locations	Warehouse spatial coordinates: Zone, Aisle, Rack, Shelf Tier, Bin, and capacity limits.	location_id (PK), bin_barcode (UQ)	None (Spatial domain entity)	B-Tree on bin_barcode, zone_code
product_batches	Perishable batch lots, manufacturing date, expiration date, on-hand qty, quarantine status.	batch_id (PK), batch_lot_number (UQ)	product_id -> products(product_id), location_id -> locations(location_id)	B-Tree on expiry_date, product_id, status (FEFO index)
purchase_orders	Digital PO tracking, supplier link, approval lifecycle, total cost, delivery deadline.	po_id (PK), po_number (UQ)	supplier_id -> suppliers(supplier_id)	B-Tree on po_number, status, order_date
goods_receipt_notes	Inbound dock inspection, PO line validation, accepted vs damaged count, digital invoice link.	grn_id (PK), grn_number (UQ)	po_id -> purchase_orders(po_id)	B-Tree on grn_number, po_id, received_at

Table Name	Primary Purpose & Operational Entity	Primary Key & Unique Keys	Key Foreign Key Constraints	Index Optimization Strategy
inventory_transactions	Immutable append-only ACID ledger recording all stock movements with row-locking support.	transaction_id (PK, BIGSERIAL)	product_id, batch_id, location_id, user_id	Composite B-Tree on (product_id, recorded_at DESC), batch_id
pos_registers	Authorized branch super shop checkout cash registers and real-time sync telemetry.	register_id (PK), terminal_mac_ip (UQ)	branch_store_id (Reference)	B-Tree on register_code, store_code, status

Below is the complete, executable Data Definition Language (DDL) conceptual script for PostgreSQL 16+ / MySQL 8.0+, demonstrating the exact schema definition, referential foreign key actions, and performance indexes:

-- RetailSync Core Relational Schema Blueprint (PostgreSQL / MySQL DDL)

```
-- =====
-- RetailSync Enterprise Relational Database Schema Blueprint (DDL)
-- Engine: PostgreSQL 16+ / MySQL 8.0+ (ACID Compliant, 3NF Normalized)
-- Course: SE-231 (Software System Analysis & Design / Capstone Project 2)
-- Author: Raisul Islam Likhon (Section: SWE-44D)
-- =====

-- 1. Suppliers Master Table
CREATE TABLE suppliers (
    supplier_id SERIAL PRIMARY KEY,
    company_name VARCHAR(150) NOT NULL,
    trade_license_no VARCHAR(80) UNIQUE,
    contact_email VARCHAR(100) NOT NULL,
    phone_number VARCHAR(25) NOT NULL,
    avg_lead_time_days NUMERIC(5,2) NOT NULL DEFAULT 7.00,
    lead_time_std_dev NUMERIC(5,2) NOT NULL DEFAULT 1.50,
    payment_terms VARCHAR(50) DEFAULT 'Net 30',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 2. Products Master Catalog
CREATE TABLE products (
    product_id SERIAL PRIMARY KEY,
    sku_code VARCHAR(50) UNIQUE NOT NULL,
    barcode_ean VARCHAR(30) UNIQUE NOT NULL,
    product_name VARCHAR(200) NOT NULL,
    category VARCHAR(50) NOT NULL CHECK (category IN ('FMCG_DRY', 'PERISHABLE_PRODUCE', 'DAIRY_CHILLED', 'FROZEN', 'HOUSEHOLD')),
    storage_temp VARCHAR(30) NOT NULL DEFAULT 'AMBIENT',
    unit_of_measure VARCHAR(20) NOT NULL DEFAULT 'UNIT',
    unit_cost NUMERIC(10,2) NOT NULL CHECK (unit_cost > 0),
    holding_cost_annual NUMERIC(10,2) NOT NULL CHECK (holding_cost_annual >= 0),
    ordering_cost_fixed NUMERIC(10,2) NOT NULL CHECK (ordering_cost_fixed >= 0),
```

```

min_shelf_life_receiving_pct NUMERIC(5,2) NOT NULL DEFAULT 75.00,
reorder_point INT NOT NULL DEFAULT 0,
safety_stock_threshold INT NOT NULL DEFAULT 0,
supplier_id INT REFERENCES suppliers(supplier_id) ON DELETE RESTRICT,
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 3. Spatial Warehouse Locations Topology
CREATE TABLE locations (
    location_id SERIAL PRIMARY KEY,
    zone_code VARCHAR(15) NOT NULL,
    aisle_number VARCHAR(10) NOT NULL,
    rack_number VARCHAR(10) NOT NULL,
    shelf_tier VARCHAR(10) NOT NULL,
    bin_barcode VARCHAR(30) UNIQUE NOT NULL,
    capacity_units INT NOT NULL DEFAULT 200,
    is_active BOOLEAN NOT NULL DEFAULT TRUE
);

-- 4. Inventory Batches (FEFO/FIFO Expiry Engine)
CREATE TABLE product_batches (
    batch_id SERIAL PRIMARY KEY,
    batch_lot_number VARCHAR(60) UNIQUE NOT NULL,
    product_id INT NOT NULL REFERENCES products(product_id) ON DELETE RESTRICT,
    location_id INT NOT NULL REFERENCES locations(location_id) ON DELETE RESTRICT,
    mfg_date DATE NOT NULL,
    expiry_date DATE NOT NULL,
    initial_quantity INT NOT NULL CHECK (initial_quantity > 0),
    current_quantity INT NOT NULL CHECK (current_quantity >= 0),
    allocated_quantity INT NOT NULL DEFAULT 0 CHECK (allocated_quantity >= 0),
    quarantine_status VARCHAR(20) NOT NULL DEFAULT 'AVAILABLE'
    CHECK (quarantine_status IN ('AVAILABLE', 'QUARANTINED', 'EXPIRED', 'DEPLETED')),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 5. Digital Purchase Orders (PO)
CREATE TABLE purchase_orders (
    po_id SERIAL PRIMARY KEY,
    po_number VARCHAR(50) UNIQUE NOT NULL,
    supplier_id INT NOT NULL REFERENCES suppliers(supplier_id) ON DELETE RESTRICT,
    total_amount NUMERIC(12,2) NOT NULL DEFAULT 0.00,
    status VARCHAR(25) NOT NULL DEFAULT 'DRAFT'
    CHECK (status IN ('DRAFT', 'ISSUED', 'PARTIAL_RECEIVED', 'COMPLETED', 'CANCELLED')),
    order_date DATE NOT NULL DEFAULT CURRENT_DATE,
    expected_delivery_date DATE NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 6. Goods Receipt Notes (GRN) Inbound Inspection
CREATE TABLE goods_receipt_notes (
    grn_id SERIAL PRIMARY KEY,
    grn_number VARCHAR(50) UNIQUE NOT NULL,
    po_id INT NOT NULL REFERENCES purchase_orders(po_id) ON DELETE RESTRICT,
    supplier_invoice_ref VARCHAR(80),
    accepted_quantity INT NOT NULL DEFAULT 0,

```



```

rejected_damaged_quantity INT NOT NULL DEFAULT 0,
receiving_clerk_id INT NOT NULL,
received_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 7. Immutable Inventory Transactions Audit Ledger (ACID Row-Locking)
CREATE TABLE inventory_transactions (
  transaction_id BIGSERIAL PRIMARY KEY,
  product_id INT NOT NULL REFERENCES products(product_id) ON DELETE RESTRICT,
  batch_id INT REFERENCES product_batches(batch_id) ON DELETE RESTRICT,
  location_id INT NOT NULL REFERENCES locations(location_id) ON DELETE RESTRICT,
  transaction_type VARCHAR(30) NOT NULL CHECK (transaction_type IN
    ('INBOUND_GRN', 'DIRECTED_PUTAWAY', 'POS_SALE_FEFO', 'STORE_DISPATCH', 'DAMAGE_QUARANTINE',
    'CYCLE_COUNT_ADJUSTMENT')),
  quantity INT NOT NULL CHECK (quantity <> 0),
  reference_document_no VARCHAR(80),
  user_id INT NOT NULL,
  recorded_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- 8. Branch POS Registers & Sync Registry
CREATE TABLE pos_registers (
  register_id SERIAL PRIMARY KEY,
  register_code VARCHAR(30) UNIQUE NOT NULL,
  store_branch_code VARCHAR(30) NOT NULL,
  terminal_ip_mac VARCHAR(60) NOT NULL,
  is_online BOOLEAN NOT NULL DEFAULT TRUE,
  last_sync_timestamp TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- Strategic B-Tree Indexes for Sub-Second Concurrency & Real-Time Telemetry
CREATE INDEX idx_products_barcode ON products(barcode_ean);
CREATE INDEX idx_products_sku ON products(sku_code);
CREATE INDEX idx_batches_fefo_queue ON product_batches(product_id, expiry_date, quarantine_status);
CREATE INDEX idx_transactions_product_date ON inventory_transactions(product_id, recorded_at DESC);
CREATE INDEX idx_transactions_batch_id ON inventory_transactions(batch_id);
CREATE INDEX idx_locations_bin_barcode ON locations(bin_barcode);

```

6.6 Stock Alert & Severity Escalation Model

To prevent alert fatigue while ensuring immediate response to operational emergencies, RetailSync establishes an explainable five-level severity model governed by deterministic rules:

Level	Designation	Trigger Conditions	System Action & Floor Workflow	Primary Stakeholders
Level 0	Normal Operating State	Stock levels between Reorder Point (ROP) and Maximum Capacity; Expiry > 90 days.	Standard operations; real-time dashboard telemetry updates; no intervention needed.	Floor Operators, Supervisors

Level	Designation	Trigger Conditions	System Action & Floor Workflow	Primary Stakeholders
Level 1	Routine Replenishment Advisory	Inventory position approaches ROP (within 10% buffer); or planned promotional demand surge.	Automated advisory generated in Procurement Queue; suggested order quantity calculated via EOQ.	Procurement Officers, Buyers
Level 2	Attention: Slow-Moving / Expiry Watch	Stock turnover rate drops below 25th percentile for category; or batch expiry window between 30 and 60 days.	Floor alert to trigger clearance allocation; automated notification to retail merchandising for promotional discounting.	Inventory Supervisor, Store Managers
Level 3	Concern: Imminent Stockout / Safety Stock Breach	Inventory Position drops strictly below Safety Stock threshold; or batch expiry within 14 days.	High-priority alert badge; automated Draft Purchase Order dispatched to approved suppliers; store order quota rationing enabled.	Procurement Officers, SCM Head
Level 4	Critical Emergency / Quarantine Lock	Physical stock reaches zero (Stockout); or batch expiration date reached; or severe physical damage detected.	Immediate system lock on SKU/batch to prevent picking; automated quarantine bin transfer order; urgent escalation alert to VP Supply Chain.	All Floor Staff, Procurement, Executives

6.7 Barcode & Handheld Data Capture Strategy

Recognizing that industrial rugged terminals cost upwards of 60,000 BDT per unit, RetailSync implements a highly accessible data capture strategy: the mobile scanning interface is built as a progressive web application (PWA) compatible with standard Android smartphones paired with low-cost Bluetooth barcode trigger grips (under 4,000 BDT) or direct camera-based barcode libraries (ZXing / Html5-QRCode). Auditory beeps and haptic vibration provide instantaneous feedback, enabling operators to scan items rapidly without continuously looking at the screen.

6.8 Hybrid Enterprise Edge Expansion: Integrating WarePulse IoT Gateways into RetailSync

A critical architectural insight emerged from our comparative evaluation with the WarePulse industrial IoT prototype: while handheld mobile scanning (PWA + Bluetooth trigger) is optimal for retail floor picking and store operations, central grocery distribution warehouses handling 50+ pallet deliveries daily benefit immensely from stationary automated dock gates. RetailSync provides native architectural extensibility to incorporate the

WarePulse edge IoT stack as an optional high-throughput inbound dock gateway without modifying the core software platform:

Architecture Tier	Component Specification	Operational Function in Central Distribution Center
Edge Hardware Station	ESP32 Dual-Core 240MHz MCU + RC522 RFID / Industrial Fixed Barcode Portal	Positioned at central warehouse inbound loading dock gates; interrogates incoming bulk pallet tags hands-free as forklifts pass through.
Edge Offline Buffering	LittleFS Non-Volatile Flash Memory Cache	Stores up to 10,000 scanned pallet events during factory Wi-Fi brownouts; guarantees zero telemetry loss during network dropouts.
Secure Telemetry Ingestion	MQTT Broker (Mosquitto/EMQX) over TLS 1.3 with X.509 Auth	Transmits dock gate scan events in sub-second JSON payloads directly to RetailSync's Inbound Ingestion Worker service.
Unified Enterprise Core	RetailSync Central Relational Core (PostgreSQL / MySQL)	Processes edge RFID/barcode dock events, automatically matches them to open Purchase Orders, and creates draft GRNs without human data entry.

Under this hybrid enterprise architecture, forklifts driving pallets through loading dock doors pass under an overhead ESP32 RFID/fixed-barcode scanner array. The edge node buffers the scan telemetry into non-volatile LittleFS flash memory during Wi-Fi brownouts and pushes structured JSON payloads via TLS-encrypted MQTT (QoS 1) directly to RetailSync's Inbound Worker service, instantly reconciling open Purchase Orders and generating draft GRNs with zero human manual intervention.

7. Intelligent System Strategy & Inventory Optimization Algorithms

A foundational differentiator of RetailSync is its integration of mathematical decision support models directly into the operational application layer. Rather than relying on black-box artificial intelligence where explainability is compromised, RetailSync combines rigorous operations research formulations with targeted unsupervised machine learning for anomaly detection.

7.1 Dynamic Economic Order Quantity (EOQ)

The EOQ model calculates the optimal order batch size (Q^*) that minimizes total annual inventory holding and ordering costs:

Mathematical Formulation: Economic Order Quantity

$$EOQ = \sqrt{(2 * D * S) / H}$$

Where:

- D = Annual Demand (units/year), dynamically extracted from rolling transactional sales history.
- S = Fixed Ordering Setup Cost per purchase order (administrative, documentation, transport inspection fees in BDT).
- H = Unit Holding Cost per year (cost of warehouse cubic footprint, refrigeration, tied-up working capital, insurance, and obsolescence risk in BDT/unit/year).

7.2 Greasley's Statistical Safety Stock Model

Standard naive safety stock heuristics assume constant supplier lead times or static demand rates. In Bangladeshi retail, however, supplier lead times fluctuate wildly due to traffic congestion, political strikes (hartals), and distributor stockouts. RetailSync implements Greasley's advanced statistical safety stock formula, which simultaneously models independent variances in both daily consumer demand and supplier delivery lead time:

Mathematical Formulation: Greasley Statistical Safety Stock

$$SS = Z * \sqrt{(L_{avg} * \sigma_d^2) + (d_{avg}^2 * \sigma_L^2)}$$

Where:

- Z = Statistical service level factor ($Z = 1.28$ for 90%, $Z = 1.65$ for 95%, $Z = 1.96$ for 97.5%, $Z = 2.33$ for 99% non-stockout probability).
- L_{avg} = Average supplier lead time in days.
- σ_L = Standard deviation of supplier lead time in days (capturing delivery volatility).
- d_{avg} = Average daily demand rate in units.
- σ_d = Standard deviation of daily demand in units (capturing retail sales volatility).

7.3 Dynamic Reorder Point (ROP)

The Reorder Point establishes the precise inventory threshold that autonomously triggers replenishment purchase orders:

Mathematical Formulation: Dynamic Reorder Point

$$ROP = (d_{avg} * L_{avg}) + SS$$

When the net Inventory Position (On-Hand Stock + Open POs - Allocated Store Orders) drops to or below ROP, the system automatically drafts a Purchase Order for the calculated EOQ batch size.

7.4 Concrete Super Shop Case Study: 1-Litre Fortified Soyabean Oil

To demonstrate the mathematical rigor of RetailSync, consider a high-velocity staple commodity in a Dhaka central retail warehouse (1-Litre Bottled Fortified Soyabean Oil):

- Annual Demand (D) = 36,500 bottles (Average daily demand $d_{avg} = 100$ bottles/day, standard deviation $\sigma_d = 20$ bottles/day).
- Order Setup Cost (S) = 1,200 BDT per purchase order (administrative processing, quality sampling, dock labor).
- Unit Holding Cost (H) = 12 BDT per bottle/year (cost of warehouse racking space, capital cost at 9%, insurance).
- Supplier Lead Time: Average lead time $L_{avg} = 6$ days, standard deviation $\sigma_L = 2$ days.
- Target Customer Service Level = 95% ($Z = 1.65$).

Step 1: Calculate Economic Order Quantity (EOQ):

$$EOQ = \sqrt{(2 * 36,500 * 1,200) / 12} = \sqrt{(87,600,000 / 12)} = \sqrt{7,300,000} = 2,702 \text{ bottles.}$$

Step 2: Calculate Greasley Statistical Safety Stock (SS):

$$\text{Term 1 (Demand Variance)} = L_{avg} * \sigma_d^2 = 6 * (20^2) = 6 * 400 = 2,400.$$

$$\text{Term 2 (Lead Time Variance)} = d_{avg}^2 * \sigma_L^2 = (100^2) * (2^2) = 10,000 * 4 = 40,000.$$

$$SS = 1.65 * \sqrt{(2,400 + 40,000)} = 1.65 * \sqrt{42,400} = 1.65 * 205.91 = 340 \text{ bottles.}$$

Step 3: Calculate Dynamic Reorder Point (ROP):

$$ROP = (100 * 6) + 340 = 600 + 340 = 940 \text{ bottles.}$$

Automated Decision Output

Managerial Decision Rule Generated by RetailSync:

"When total net inventory position falls to 940 bottles, the system autonomously flags the SKU and generates a draft Purchase Order for 2,702 bottles, guaranteeing a 95% statistical assurance that stock will not deplete prior to supplier delivery."

7.5 Machine Learning Inventory Anomaly & Shrinkage Detection

In addition to mathematical replenishment, RetailSync integrates an unsupervised Machine Learning model (Isolation Forest) trained on cycle count historical variances. The algorithm flags suspicious shrinkage anomalies where stock discrepancies deviate significantly from normal baseline breakages, correlating discrepancies across specific shifts, operators, and high-value merchandise to detect pilferage clusters proactively.

PART C

TARGET USERS & REAL-LIFE SCENARIOS

8. Target Users, Personas and Stakeholder Analysis

A warehouse management system succeeds only when its user interfaces are tailored to the physical environments and cognitive demands of its diverse user base. RetailSync serves primary frontline retail and warehouse management user groups:

8.1 Target Users Identification

User Category	Operational Role	Primary Daily Responsibilities & Pain Points	Key RetailSync Capabilities Utilized
Warehouse Floor Operator	Direct Operational (Floor)	Pallet unloading, bin putaway, shelf picking, order packing. Pain: searching for misplaced boxes, heavy paperwork.	Mobile scanner UI, directed putaway prompts, digital pick lists, instant barcode verification.
Inbound Receiving Clerk	Tactical Operational (Dock)	Inspecting supplier deliveries, validating purchase orders, logging damages. Pain: paper PO matching errors.	Digital PO lookup, barcode printing, discrepancy logging, automated Goods Receipt Note (GRN) generation.
Inventory Floor Supervisor	Tactical Management	Stock accuracy, bin space utilization, cycle counting, shelf life tracking. Pain: phantom inventory, unrecorded shrinkage.	Live 2D spatial bin monitor, cycle count assignment, expiry tracking dashboard, discrepancy reconciliation.
Procurement & SCM Officer	Strategic Tactical	Vendor negotiation, purchase order issuance, stock replenishment planning. Pain: unexpected stockouts, volatile supplier lead times.	Algorithmic replenishment portal (EOQ/ROP), supplier SLA scorecard, automated reorder triggers, lead-time variance tracking.
Branch Super Shop Store Manager	Operational Recipient (Store)	Ordering store stock from central warehouse, receiving store deliveries, shelf restocking. Pain: unfulfilled requisitions, stockout shelves.	Store Requisition Portal, live central DC inventory visibility, dispatch delivery tracking, transit loss reconciliation.

User Category	Operational Role	Primary Daily Responsibilities & Pain Points	Key RetailSync Capabilities Utilized
Warehouse Systems Admin / DevOps	Technical Support	User access provisioning, system uptime, database performance, audit trail integrity. Pain: database locks, unauthorized changes.	Role-Based Access Control (RBAC), database indexing telemetry, system health monitors, immutable audit logs.
Executive Management (Director/VP)	Strategic Executive	Working capital efficiency, inventory turnover, perishable waste reduction, retail chain profitability. Pain: lack of high-level insights.	Executive KPI dashboard, inventory turnover analytics, shrinkage reports, gross margin return on inventory (GMROI).

8.2 Core Operational Personas (From Comprehensive Study)

The operational study defines three fundamental personas who interact daily with the core RetailSync ecosystem:

User Persona	Role Description	System Needs
Store Manager	Oversees supply chain and profitability.	Needs automated PO drafts, shrinkage reports, and a macro-view of inventory health.
Cashier	Processes direct consumer sales.	Requires a minimalist interface. The system must never freeze during a scan, even if the internet drops momentarily.
Inventory Clerk	Handles loading docks and physical counting.	Needs fast, barcode-compatible interfaces for receiving large shipments accurately.

8.3 Detailed Field User Personas

To further anchor software design decisions in genuine human context, three detailed field user personas were established:

- **Persona 1 — Tariqul Islam (Warehouse Floor Manager):** Age 34, manages 18 floor staff at a central grocery distribution warehouse in Tejgaon, Dhaka. Pain points: spends 2 to 3 hours daily resolving lost pallet discrepancies and paper tally errors. RetailSync Goal: Needs a real-time 2D spatial bin map and instant mobile barcode verification to eliminate physical search latency.
- **Persona 2 — Salma Akter (Senior Procurement Executive):** Age 29, manages fresh food and FMCG vendor procurement for a 14-store retail chain. Pain points: struggles with unexpected supplier delivery delays and sudden out-of-stock weekend crises on dairy products. RetailSync Goal: Needs automated EOQ/ROP replenishment triggers and supplier lead-time variance scorecards to order accurately.

- **Persona 3 — Kamal Hossain (Branch Super Shop Store Manager):** Age 41, manages a flagship super shop outlet in Dhanmondi, Dhaka. Pain points: frequently receives incorrect goods quantities or near-expiry dairy batches from the central warehouse with zero advance notice. RetailSync Goal: Needs a clean Store Requisition Portal with live central stock visibility and automated FEFO dispatch verification.

8.4 Stakeholder Analysis Matrix

The stakeholder matrix below outlines the interest, influence, and engagement strategy for all direct and indirect project stakeholders:

Stakeholder Group	Primary Role	Core Interests & Strategic Drivers	Influence	Engagement Strategy
Warehouse Floor Staff	Direct System Users	Ease of use, fast scanning, elimination of manual paperwork, fair workload tracking.	High	Co-design mobile scanner interfaces, conduct hands-on usability testing, iterative feedback.
Procurement & Supply Chain Team	Operational Decision Makers	High service levels (>98%), low holding costs, reliable supplier lead-time tracking, automated EOQ/ROP.	High	Collaborate on replenishment algorithms, validate supplier SLA metrics, bi-weekly reviews.
Super Shop Branch Managers	Downstream Beneficiaries	On-time full delivery (OTIF), accurate store consignments, zero damaged/expired goods received.	High	Design intuitive store requisition portal, provide clear in-transit order tracking.
Executive Committee / Sponsors	Strategic Steering	Working capital reduction, waste minimization, gross margin expansion, return on investment (ROI).	High	Executive KPI dashboards, sprint demos, comprehensive financial impact reporting.
Suppliers & Distributors	External Partners	Predictable purchase orders, fast dock turnaround, clear rejection rationale on damaged lots.	Medium	Standardized electronic POs, transparent digital GRN inspection notes, SLA scorecards.
Regulatory Bodies (BFSA, BSTI)	Compliance & Legal	Consumer food safety, strict expiration date adherence, batch traceability, hygienic storage.	Medium	Enforce strict FEFO controls, immutable batch audit logs, automated compliance reporting.

8.5 Stakeholder RACI Responsibility Assignment Matrix

To establish unequivocal operational ownership and cross-functional governance throughout the capstone engineering lifecycle, the formal RACI model below delineates stakeholder roles across eleven mission-critical delivery milestones:

Project Lifecycle Phase / Core Deliverable	Floor Staff / Cashiers	Floor Supervisors	Procurement Officers	Store Managers	DevOps / SysAdmin	Developer Team	Academic Committee
Requirements Elicitation & Domain Study	Consulted	Consulted	Consulted	Consulted	Informed	Responsible / Accountable	Informed / Approver
Relational 3NF Database Schema & DDL	Informed	Consulted	Informed	Informed	Consulted	Responsible / Accountable	Consulted
ACID Concurrency & Row-Locking Engine	Informed	Informed	Informed	Informed	Consulted	Responsible / Accountable	Informed
Sub-2.0s POS Sync & Real-Time API	Consulted	Informed	Informed	Consulted	Consulted	Responsible / Accountable	Informed
FEFO/FIFO Dynamic Queue Algorithm	Informed	Consulted	Consulted	Consulted	Informed	Responsible / Accountable	Consulted
Greasley Safety Stock & EOQ Engine	Informed	Consulted	Responsible	Consulted	Informed	Accountable	Informed
Mobile Barcode Scanning PWA Interface	Responsible	Responsible	Informed	Informed	Informed	Accountable	Informed

Project Lifecycle Phase / Core Deliverable	Floor Staff / Cashiers	Floor Supervisors	Procurement Officers	Store Managers	DevOps / SysAdmin	Developer Team	Academic Committee
Immutable Audit Ledger & Security	Informed	Consulted	Consulted	Informed	Responsible	Accountable	Informed
Dual-Level Black-Box & White-Box QA	Consulted	Consulted	Consulted	Consulted	Consulted	Responsible / Accountable	Consulted / Evaluator
Final Manuscript, System Demo & Defense	Informed	Informed	Informed	Informed	Informed	Responsible / Accountable	Accountable / Approver

RACI Legend: Responsible (R) = Executes the delivery; Accountable (A) = Final decision and sign-off authority; Consulted (C) = Provides bi-directional domain input; Informed (I) = Kept updated on progress and outcomes.

9. Real-Life Operational Use-Case Scenarios

9.1 Core Real-Life Retail Scenarios

RetailSync is designed to withstand extreme real-world operating conditions common in Bangladeshi super shops:

Scenario Title	Physical Reality	RetailSync Action
The Festival Rush	During Ramadan, sugar and oil sales quadruple. A cashier scans items rapidly.	RetailSync utilizes optimistic concurrency control in the database, allowing multiple cashiers to sell sugar simultaneously without locking the system, updating the central database seamlessly.
The Expired Goods Audit	A health inspector visits the store.	The manager opens RetailSync's Expiry Dashboard, proving that the system actively tracks and flags items 7 days before expiry, ensuring no expired goods are on shelves.

Scenario Title	Physical Reality	RetailSync Action
The Network Drop	The store's ISP connection fails for 5 minutes.	The POS continues to scan items locally. Once the connection is restored, it bulk-syncs the transaction payload to the central RetailSync database (Resilience feature).

9.2 Advanced Logistics & Warehouse Scenarios

The following operational scenarios illustrate how warehouse operators and procurement officers interact with RetailSync:

- Scenario 1: High-Velocity FMCG Replenishment under Volatile Lead Times:** During a Ramadan promotional surge, daily demand for 1L edible oil spikes. The dynamic ROP algorithm detects that the remaining inventory position (850 units) has breached the reorder threshold (940 units). The system autonomously drafts a Purchase Order for 2,702 units (EOQ) assigned to the primary supplier, incorporating the supplier's 6-day lead-time variance. The procurement officer reviews and approves the PO in one click, preventing an impending stockout.
- Scenario 2: Perishable Dairy Expiry Management with Automated FEFO Allocation:** A supplier delivers 500 crates of pasteurized milk. The receiving clerk scans the supplier barcode, recording a manufacturing date of Sept 26 and expiration date of Oct 3. The system verifies that the remaining shelf life exceeds the 75% threshold and generates internal batch barcode labels. When two branch stores order milk three days later, the system's FEFO engine directs pickers strictly to this batch rather than a newer batch arriving on Sept 29, eliminating spoiled stock.
- Scenario 3: Receiving Dock Discrepancy, Damaged Goods, and Automated GRN:** A biscuit supplier delivers 200 cartons against an open PO of 200 cartons. During dock barcode scanning, the clerk discovers that 25 cartons suffered severe water and crushing damage during transit. The clerk scans the barcode, inputs 'Rejected: 25 - Damaged Packaging', and snaps a photo. The system generates a GRN for 175 accepted cartons, automatically updates the PO status to 'Partially Received', and drafts a digital credit note advisory for the 25 rejected cartons.
- Scenario 4: Multi-Store Outbound Wave Picking and Branch Staging:** Five retail branches submit morning requisitions for 40 common FMCG items. Instead of sending five pickers through the warehouse aisles independently, RetailSync consolidates the orders into a single Wave Pick. The system generates an optimized serpentine pick path guiding a single operator to collect total quantities in one pass. At the outbound staging dock, goods are sorted into branch-specific rolling cages, reducing total picker travel distance by 64%.

PART D**GOALS, SCOPE AND REQUIREMENTS****10. Project Objectives****10.1 Primary (General) Objective**

To design, architect, implement, and evaluate an automated, web-based Centralized Super Shop Warehouse Management System (RetailSync) that digitizes physical warehouse movements via handheld barcode scanning, synchronizes warehouse ledgers with frontline retail POS registers, enforces strict FEFO/FIFO batch tracking for perishable foods, optimizes inventory carrying costs and replenishment timing through mathematical decision models (EOQ/ROP), and orchestrates seamless multi-branch retail store fulfillment.

10.2 S.M.A.R.T. Specific Objectives

The project's general objective is operationalized into six Specific, Measurable, Achievable, Relevant, and Time-bound (S.M.A.R.T.) objectives:

ID	Objective Statement	Target Metric & Verifiable Evaluation Criteria	Timeline
O-01	Accelerate Inbound Receiving and Digital GRN Generation	Reduce average inbound unloading-to-dock verification time from 45 minutes to under 12 minutes per shipment (73% reduction); achieve 100% digital GRN generation with zero paper manifests.	Phase 2 (Sprint 2)
O-02	Eliminate Misplaced Stock via Directed Spatial Putaway	Achieve 99.5% bin location accuracy across all warehouse zones; reduce putaway search and placement time by 60% through barcode-guided bin destination routing.	Phase 3 (Sprint 3)
O-03	Minimize Perishable Spoilage via Automated FEFO Allocation	Achieve 100% compliance with First-Expired, First-Out (FEFO) picking logic on perishable and dairy SKUs, reducing warehouse expiry wastage by at least 65% in pilot trials.	Phase 3 (Sprint 4)

ID	Objective Statement	Target Metric & Verifiable Evaluation Criteria	Timeline
O-04	Optimize Inventory Replenishment with Mathematical DSS	Deploy dynamic EOQ, Greasley Statistical Safety Stock, and ROP models, cutting stockouts on top-tier (Category A) FMCG items by 80% while reducing excess holding stock by 22%.	Phase 4 (Sprint 4)
O-05	Streamline Outbound Multi-Store Picking and Dispatch	Increase order picking throughput from 40 lines/hour to over 95 lines/hour using consolidated wave picking; eliminate store shipping discrepancy rate to <0.3%.	Phase 4 (Sprint 5)
O-06	Enforce Complete Auditability, RBAC, and Sub-Second Telemetry	Ensure all stock transactions are recorded in an immutable ACID ledger with complete user attribution; achieve sub-1.2 second average page load and scan latency across all modules.	Phase 5 (Sprint 6)

11. Scope and Boundaries

To ensure successful completion within the rigorous academic capstone semester, the project establishes explicit functional boundaries:

11.1 Initial Capstone MVP (In-Scope)

- **Authentication & RBAC:** Role-Based Access Control (RBAC) with secure JWT authentication isolating Cashier, Clerk, Supervisor, and Manager views.
- **Product Catalog:** Product Master catalog with SKU taxonomy, ABC classification, temperature classes, and barcode generation.
- **Supplier & PO:** Supplier directory, SLA lead-time variance tracking, and digital Purchase Order workflow.
- **Inbound & GRN:** Dock receiving, barcode PO verification, damaged item recording, and automated digital GRN generation.
- **Spatial Bin Engine:** Hierarchical Zone-Aisle-Rack-Shelf-Bin topology mapping with capacity-aware directed putaway.
- **Inventory & FEFO:** ACID append-only transaction ledger, batch expiry tracking, and strict FEFO pick queue enforcement.
- **POS Synchronization:** POS Sync Simulation API executing sub-2.0s atomic stock deductions based on FIFO/FEFO batch logic.
- **Replenishment DSS:** Dynamic Economic Order Quantity (EOQ), Greasley Statistical Safety Stock, and Dynamic ROP alert triggers.

- **Outbound & Store Portal:** Multi-branch store requisition portal, wave picking consolidation, and digital delivery chalan generation.
- **Cycle Counting & Audit:** Continuous cycle counting schedules, blind physical count entry, discrepancy variance auditing, and write-offs.
- **Executive Analytics:** Real-time floor telemetry, stockout risk heatmaps, supplier SLA scorecards, and PDF/Excel reporting.

11.2 Planned Enhancements (Time Permitting)

- Unsupervised machine learning (Isolation Forest) for automatic inventory shrinkage and pilferage cluster detection.
- 2D interactive visual canvas rendering real-time heatmaps of warehouse bin utilization.
- SMS and WhatsApp automated alerts for high-priority stockout and expiry escalations.

11.3 Future Scope (Outside Capstone Scope)

- Physical automated guided vehicles (AGVs), robotic gantry cranes, and motorized conveyor sorters.
- Ultra-High Frequency (UHF) multi-meter warehouse radar triangulation and automated drone inventory counting.
- Nationwide Electronic Data Interchange (EDI) integration with enterprise legacy ERP networks.

12. Project Boundary, Safety, Security and Compliance Framework

12.1 The System Will and Will Not

- The system WILL automate warehouse stock verification, track batch expiration dates, allocate pick lists via FEFO, synchronize POS stock deductions in sub-2s, calculate mathematical replenishment parameters, and manage store requisitions.
- The system WILL NOT perform financial accounting general ledger balancing, process end-customer consumer retail payments, replace human physical forklift operations, or manufacture custom scanning hardware.

12.2 Security, Integrity, and Audit Framework

All data transmissions are encrypted using TLS 1.3. User passwords are encrypted using bcrypt / Argon2id hashing with unique salts. All inventory balance changes are enforced server-side inside atomic database transactions (ACID compliance) with row-level locking (SELECT ... FOR UPDATE) to guarantee that concurrent sales and picking operations never produce negative inventory balances. Every stock modification, write-off, and permission change is permanently logged in an immutable, tamper-evident audit ledger.

12.3 Regulatory and Industry Compliance in Bangladesh

RetailSync is explicitly architected to satisfy national legal and regulatory requirements in Bangladesh: 1. Bangladesh Food Safety Act 2013: Requires strict traceability of food origins and mandates that expired food items be permanently segregated. RetailSync enforces automated quarantine locks that mechanically prevent any batch within 48 hours of expiration from being picked. 2. BSTI Standards: Ensures compliance with statutory packaged food labeling and expiration date formats. 3. Personal Data Protection Ordinance 2025 (Ordinance No. 61 of 2025): Ensures that employee and supplier personal contact data is encrypted, access-restricted, and strictly protected against unauthorized disclosure.

13. Requirements Analysis Summary

13.1 Core Functional Requirements (From Comprehensive Study)

The foundational functional requirements governing core supermarket operations are summarized below:

ID	Module	Specific Requirement	Priority
FR-01	Auth	System shall enforce RBAC isolating Cashier, Clerk, and Manager views.	Must
FR-10	Inventory	System shall track Product ID, SKU, Batch Number, and Expiry Date.	Must
FR-11	Inventory	System shall allow status tagging: Available, Quarantined, Sold.	Must
FR-20	Receiving	System shall generate a Goods Receipt Note (GRN) linked to a specific PO.	Must
FR-21	Receiving	Clerks can flag received items as 'Damaged' upon intake.	Should
FR-30	POS Sync	API shall deduct inventory based on strict FIFO/FEFO batch logic.	Must
FR-40	Alerts	System shall allow Managers to configure a Reorder Point (ROP) per SKU.	Must
FR-41	Alerts	System shall flag batches expiring within X days (configurable).	Must
FR-50	Auditing	Every stock modification must log the timestamp, User ID, and old/new value.	Must

13.2 Summary of Functional Requirements by Module (M-01 to M-11)

The complete functional requirements taxonomy encompasses 11 modules and 78 detailed specifications prioritized using MoSCoW. (The exhaustive specification is presented in Appendix A):

Module ID & Name	Requirement ID Range	Core Functional Capabilities Covered	MoSCoW Priority
M-01: Authentication, RBAC & Profile	FR-01 to FR-06	Secure login, JWT tokens, RBAC roles (Operator, Clerk, Supervisor, Procurement, Store Manager, Admin), password reset, session audit.	Must Have (MVP Core)
M-02: Product Master & Hierarchy	FR-07 to FR-13	SKU management, barcode assignment, category hierarchy (FMCG, Perishable, Chilled, Dry), temperature requirements, shelf-life rules.	Must Have (MVP Core)
M-03: Supplier & Purchase Orders	FR-14 to FR-20	Supplier directory, lead-time variance tracking, digital Purchase Order generation, approval workflows, PO status lifecycle.	Must Have (MVP Core)
M-04: Inbound Receiving & Digital GRN	FR-21 to FR-28	Dock receiving, barcode scan verification against PO, damaged item logging, digital GRN generation, credit note flagging.	Must Have (MVP Core)
M-05: Spatial Bin & Putaway Engine	FR-29 to FR-35	Zone-Aisle-Rack-Shelf-Bin 2D mapping, capacity constraints, directed putaway suggestions based on SKU velocity and product class.	Must Have (MVP Core)
M-06: Real-Time Ledger & FEFO Engine	FR-36 to FR-44	Double-entry inventory ledger, batch/lot tracking, expiry date monitoring, FEFO priority picking queue, automated quarantine lock.	Must Have (MVP Core)
M-07: Replenishment & Decision Support	FR-45 to FR-52	Dynamic EOQ calculator, Greasley's Safety Stock with service levels (90-99%), dynamic ROP alerts, automated PO draft creation.	Must Have (MVP Core)

Module ID & Name	Requirement ID Range	Core Functional Capabilities Covered	MoSCoW Priority
M-08: Outbound Store Wave Picking	FR-53 to FR-60	Multi-store requisition ingestion, wave creation, shortest-path digital pick-lists, pick verification scanning, staging, dispatch note.	Must Have (MVP Core)
M-09: Cycle Counting & Shrinkage Audit	FR-61 to FR-66	ABC-classified cycle counting schedules, blind physical count entry, discrepancy variance analysis, stock write-off approvals.	Should Have / Must
M-10: Reporting, Dashboards & Analytics	FR-67 to FR-72	Real-time floor telemetry, stockout risk heatmaps, supplier SLA scorecards, inventory turnover & GMROI metrics, PDF/Excel export.	Should Have
M-11: Security, Audit Trail & Compliance	FR-73 to FR-78	Immutable audit log for all stock movements, Bangladesh Food Safety Act compliance reports, PDPO 2025 privacy compliance.	Must Have (Cross-Cutting)

13.3 Non-Functional Requirements & Performance Metrics

The key quantifiable performance and security metrics established in the operational study are detailed below:

ID	Category	Specific Metric
NFR-01	Latency	The POS deduction API must respond in under 2.0 seconds under normal load.
NFR-02	Data Integrity	Database must use transaction blocks to prevent partial updates if a crash occurs.
NFR-03	Security	All passwords must be hashed (bcrypt/Argon2id). No plaintext credentials stored.
NFR-04	Concurrency	System must handle at least 5 simultaneous POS registers deducting stock.
NFR-05	Auditability	Audit logs must be immutable (append-only) via database-level restrictions.

The complete suite of fourteen Non-Functional Requirements (NFR-01 to NFR-14) governing enterprise reliability is presented below:

ID	Category	Requirement Statement & Verification Metric	Priority
NFR-01	Security & Encryption	All web and mobile traffic shall be encrypted using TLS 1.3. Sensitive database credentials and session tokens shall be encrypted at rest using AES-256.	Must Have
NFR-02	Authentication & Passwords	User passwords shall be hashed using Argon2id or bcrypt (work factor ≥ 12). Tokens shall expire within 8 hours with automatic refresh token rotation.	Must Have
NFR-03	Data Integrity & ACID	All inventory state transitions (receiving, transfer, picking, adjustment) shall execute inside atomic database transactions, guaranteeing zero negative balances.	Must Have
NFR-04	System Performance	Barcodes scanned via handheld readers or web UI shall be verified and processed within ≤ 350 milliseconds. Dashboard analytics pages shall render in ≤ 1.5 seconds.	Must Have
NFR-05	Concurrent Workload	The system shall support at least 50 concurrent active warehouse scanners and web dashboard sessions without performance degradation or database locking.	Must Have
NFR-06	High Availability	The backend services and database shall maintain 99.8% operational uptime during warehouse working hours (6:00 AM to 11:00 PM BST).	Must Have

ID	Category	Requirement Statement & Verification Metric	Priority
NFR-07	Fault Tolerance & Recovery	In case of server crash or power failure, database Point-In-Time Recovery (PITR) shall ensure a Recovery Point Objective (RPO) ≤ 1 minute and RTO ≤ 15 minutes.	Must Have
NFR-08	Usability & Ergonomics	The mobile scanning UI shall be optimized for one-handed operation on 5.5-inch rugged smartphones with high-contrast text and audio-haptic feedback.	Must Have
NFR-09	Mobile Responsiveness	All operator workflows shall render seamlessly across desktop management consoles (1920x1080) and ruggedized Android barcode terminals (720x1280).	Must Have
NFR-10	Maintainability & Modularity	The codebase shall follow clean layered architecture with documented RESTful API endpoints and unit test coverage $\geq 80\%$ for core algorithms.	Must Have
NFR-11	Scalability	The database schema and indexing strategy shall support up to 50,000 SKUs, 500,000 active inventory batches, and 5,000,000 transaction records without query lag.	Should Have
NFR-12	Auditability & Logging	Every stock adjustment, manual override, and permission change shall be logged with exact user ID, timestamp, IP address, and previous/new state.	Must Have

ID	Category	Requirement Statement & Verification Metric	Priority
NFR-13	Compliance & Safety	Batch expiration tracking shall conform to Bangladesh Food Safety Act 2013 and BSTI standards, preventing any expired item from being dispatched.	Must Have
NFR-14	Cost-Efficiency	The system shall run on modern open-source software stacks (Node.js/Next.js, PostgreSQL, Docker) to maintain low total cost of ownership (TCO) for retail chains.	Must Have

13.4 Feature-to-Problem Mapping

To verify complete problem coverage, every core RetailSync feature is explicitly linked back to the problems defined in Section 3:

Core System Feature	Operational Real-Life Problem Resolved	Delivered Business Value	Problem Link
Handheld Barcode Receiving	Unchecked supplier errors, manual paper manifests	Instant verification, zero PO mismatch	P-01
Directed Spatial Putaway	Disorganized racks, long search times	60% reduction in putaway and picking travel latency	P-02
Strict FEFO Expiry Allocation	Severe food spoilage from picking newer stock first	Elimination of expired stock dispatch, 65% waste cut	P-03
Continuous Cycle Counting	Phantom inventory, undetected shrinkage and theft	99.5% inventory record accuracy, early theft detection	P-04
Algorithmic EOQ & ROP Engine	Overstocking capital lockup and weekend stockouts	Balanced working capital, 80% reduction in stockouts	P-05
Consolidated Wave Picking	Pickers walking aisles repeatedly for separate orders	Doubled picking throughput, faster store dispatches	P-06
Supplier SLA Scorecards	Zero visibility into supplier lead-time volatility	Data-backed vendor negotiation and safety stock tuning	P-07

Core System Feature	Operational Real-Life Problem Resolved	Delivered Business Value	Problem Link
Store Requisition Portal	Phone/email ordering, central warehouse stock silos	Transparent stock allocation, zero duplicate orders	P-08
Immutable Audit Ledger	Lack of accountability, food safety violations	Complete chain-of-custody, BFSA compliance guarantee	P-09

PART E

HOW WE WILL BUILD IT

14. Feasibility Analysis

14.1 5-Dimension Feasibility Assessment

A rigorous feasibility assessment across five critical dimensions confirms that RetailSync can be delivered within the academic semester:

- **1. Technical Feasibility:** Fully feasible utilizing mature, open-source enterprise stacks (Node.js/Next.js, Python/FastAPI, PostgreSQL 16 / MySQL, Docker). Scanning leverages ubiquitous standard barcode hardware and mobile browser APIs without requiring custom silicon or proprietary firmware.
- **2. Schedule Feasibility:** Feasible within a 14-week academic semester by implementing a strict Agile Scrum methodology across six 2-week iterations. Core receiving, putaway, POS sync, and FEFO picking are delivered in early sprints, ensuring a working MVP is demonstrable by mid-term.
- **3. Economic Feasibility:** Highly cost-effective. Built exclusively on free, open-source software, eliminating licensing fees. Development and testing utilize personal computers, smartphones, and student cloud hosting credits (Vercel, Supabase), achieving near-zero capital expenditure.
- **4. Operational Feasibility:** High user adoption potential. The mobile scanning UI is designed for simple, one-handed operation with high-contrast text and dual Bangla-English language support, minimizing training requirements for floor staff.
- **5. Legal & Ethical Feasibility:** Fully compliant with the Bangladesh Food Safety Act 2013 and the Personal Data Protection Ordinance 2025. Uses synthetic inventory datasets for academic testing, avoiding sensitive real-world trade secrets.

14.2 What Is Implemented Now vs Future Scope Matrix

Functional Capability	Available Industrial Tech	Included in Capstone MVP	Deferred to Future Scope
Inbound Barcode Scans	Handheld Android terminals, 2D Barcode scanners	Web-based camera scan + USB/Bluetooth barcode scanner integration	Automated high-speed conveyor tunnel scanners
Spatial Bin Putaway	2D/3D digital twin warehouse maps, pick-to-light	Visual 2D Zone-Aisle-Rack-Bin layout with capacity validation	Automated Guided Vehicles (AGVs) and robotic shuttle putaway
FEFO Expiry Allocation	Relational database date indexing, dynamic queues	Native PostgreSQL date indexing with automated quarantine locks	Automated RFID sensor tags measuring real-time produce decay
Replenishment Decision	Statistical engines, complex enterprise ERP	Dynamic EOQ, Greasley Statistical Safety Stock, and ROP triggers	Deep Reinforcement Learning multi-echelon neural networks
Multi-Store Wave Picking	Voice picking, automated pick-to-light carts	Consolidated wave creation, shortest-path digital pick-lists	Autonomous robotic mobile picking arms

Functional Capability	Available Industrial Tech	Included in Capstone MVP	Deferred to Future Scope
Cycle Count Discrepancy	Cycle counting software, RFID drone flyover	Blind cycle count entry, automated variance calculation	Autonomous drone inventory scanning across vertical racks
Store Requisition Portal	B2B web portals, electronic data interchange	Responsive web portal for branch managers to request stock	Nationwide retail Electronic Data Interchange (EDI) network

15. Development Methodology

15.1 Comparative SDLC Methodology Evaluation Matrix

Selecting the optimal Software Development Life Cycle (SDLC) is a foundational software engineering decision in SE-231. The candidate methodologies are rigorously evaluated below against the specific technical realities of a real-time retail WMS:

SDLC Model	Core Engineering Strengths	Critical Weaknesses for Retail WMS	Capstone Suitability Verdict
Waterfall Model	Predictable linear stages, thorough upfront documentation, clear milestone stage gates.	Completely rigid; assumes 100% frozen requirements upfront; fails to accommodate unexpected database concurrency conflicts or floor usability feedback discovered during testing.	Unsuitable
V-Model (Verification & Validation)	Exceptional verification discipline; rigorous test-case traceability for every development phase.	Retains Waterfall's rigidity; changes in POS API integration or barcode scan ergonomics late in the cycle require restarting requirements.	Partially Suitable
Spiral Model	Deep risk analysis at every iteration; well suited for high-budget, high-uncertainty industrial projects.	Excessive managerial overhead, heavy documentation burden, and complex milestone tracking impractical for a 14-week university capstone.	Over-Engineered / Unsuitable

SDLC Model	Core Engineering Strengths	Critical Weaknesses for Retail WMS	Capstone Suitability Verdict
Agile Scrum with Evolutionary Prototyping	Rapid 2-week iterations, continuous stakeholder feedback, functional working MVP (schema + API) delivered in early sprint, continuous concurrency stress-testing, and rapid UI adaptation.	Requires disciplined sprint timeboxing and active backlog management to prevent scope creep into hardware or robotics.	Highly Recommended (Selected Model)

15.2 In-Depth Justification for Agile Scrum with Evolutionary Prototyping

Agile Scrum with Evolutionary Prototyping is conclusively selected based on four empirical software engineering realities:

- **1. Empirical Concurrency Validation:** Real-Time Concurrency Discovery: Transactional concurrency locks (SELECT ... FOR UPDATE) and sub-2.0s POS deduction latencies cannot be verified through abstract theoretical paperwork. Empirical stress testing across simulated multi-register checkouts in early sprints is essential to identify and eliminate database deadlocks.
- **2. Early Working Core Demonstration:** Early Working Core MVP: By the conclusion of Sprint 2 (Week 4), the team delivers a functioning relational database schema and working RESTful ingestion API, guaranteeing tangible software progress for academic milestones and supervisor reviews.
- **3. Rapid Ergonomic UI Adaptation:** Continuous Usability & Barcode Feedback: Floor operator ergonomics (e.g. one-handed mobile scanning, high-contrast alerts in warehouse lighting, auditory scan feedback) require continuous hands-on user interface iterations rather than end-of-project testing.
- **4. Decoupled Parallel Track Execution:** Parallel Full-Stack Development: The decoupled 4-tier architecture allows backend API engineering and React frontend development to advance synchronously against pre-agreed OpenAPI/Swagger schema contracts without blocking dependencies.

15.3 Capstone Sprints & Incremental Delivery Plan

The project execution is mapped into six 2-week sprints matching the 14-week university semester timeline:

Iteration	Sprint Focus	Requirements Addressed	Sprint Deliverable & Milestone
Sprint 1 (Weeks 1-3)	Architecture & Foundation	FR-01 to FR-06, FR-07 to FR-13, NFR-01 to NFR-03	Core system architecture, normalized PostgreSQL schema, JWT authentication, RBAC, Product Master catalog.
Sprint 2 (Weeks 3-5)	Supplier & Inbound Receiving	FR-14 to FR-28, NFR-04, NFR-08	Supplier directory, PO workflow, mobile dock receiving module, barcode scan verification, digital GRN generator.

Iteration	Sprint Focus	Requirements Addressed	Sprint Deliverable & Milestone
Sprint 3 (Weeks 5-8)	Spatial Putaway & FEFO Ledger	FR-29 to FR-44, NFR-03, NFR-09	Warehouse spatial bin mapping, directed putaway engine, immutable transaction ledger, FEFO allocation queue.
Sprint 4 (Weeks 8-10)	Replenishment DSS Engine	FR-45 to FR-52, NFR-10, NFR-14	Dynamic EOQ, Greasley's Statistical Safety Stock, dynamic ROP alerting, automated draft PO creation module.
Sprint 5 (Weeks 10-12)	Wave Picking & Store Portal	FR-53 to FR-60, FR-61 to FR-66, NFR-05	Multi-store requisition portal, wave picking engine, pick-path optimization, cycle count auditing, dispatch chalan.
Sprint 6 (Weeks 12-14)	Analytics, Hardening & Evaluation	FR-67 to FR-78, All NFRs	Executive KPI dashboard, security hardening, user acceptance testing (UAT), final report manuscript and capstone defense.

16. Resource, Time and Budget Planning

16.1 Development & Infrastructure Resources Planning

Resource Category	Specific Technology / Asset	Purpose in Project Implementation	Acquisition & Budget Strategy
Development Framework	Next.js (React), Node.js / Express or Python Fastify	Responsive frontend web portal and high-throughput REST API services.	Open source; zero licensing cost.
Database & Caching	PostgreSQL 16 with B-Tree indexes, Redis for caching	ACID-compliant relational ledger, spatial bin tracking, and session cache.	Open source; locally hosted / Docker.
Barcode Scanning	Bluetooth 1D/2D Handheld Barcode Scanner + Smartphone Camera	Physical barcode interrogation for inbound receiving, putaway, and picking.	Existing personal devices & low-cost USB barcode reader.
Containerization	Docker & Docker Compose	Reproducible development, local multi-service testing, and deployment.	Free community edition.

Resource Category	Specific Technology / Asset	Purpose in Project Implementation	Acquisition & Budget Strategy
Cloud & Hosting	Vercel / Supabase / Render / Academic Cloud Tier	Staging environment for remote testing and capstone committee evaluation.	Free academic student developer tier.
Testing & CI/CD	Jest, Supertest, GitHub Actions	Automated unit testing of mathematical models and continuous integration.	Free open-source tooling.
Documentation & Diagrams	Mermaid.js, Draw.io, MS Word / LibreOffice	UML diagrams, flowcharts, ERD schemas, and academic project manuscripts.	Included in academic software suite.

16.2 14-Week Capstone Implementation Roadmap

The project schedule is rigorously mapped across 14 academic weeks according to the operational study roadmap:

Academic Timeframe	Core Engineering Modules & Deliverables	Key Technical Milestones
Weeks 1-3	Database Normalization (MySQL/PostgreSQL), ERD design, and API scaffolding (Python/Java).	Relational schema definition, 3NF normalization, baseline REST endpoints.
Weeks 4-6	Inventory CRUD, Batch Management, and Expiry Logic implementation.	Batch tracking, dynamic shelf-life calculations, FEFO/FIFO queue logic.
Weeks 7-9	Supplier module, PO generation, and GRN workflows.	Supplier SLA records, digital PO lifecycle, dock inspection & GRN generation.
Weeks 10-11	POS Sync Simulation (building the checkout deduction logic and FIFO handling).	Real-time POS API, atomic stock deduction, concurrency locking (<2s latency).
Weeks 12-13	Rule engines for low-stock alerts, reporting dashboards, and audit logs.	Automated ROP triggers, draft PO generation, immutable audit logging.
Week 14	System testing, concurrency stress testing, and final defense preparation.	Stress testing (5+ concurrent POS), UAT validation, capstone presentation & defense.

16.3 Budget Approach

The project follows a frugal engineering approach: development runs entirely on student hardware and open-source stacks. Production demonstration will utilize free-tier cloud hosting (Vercel for frontend, Supabase/Render for managed PostgreSQL). Estimated out-of-pocket expenditure is limited to approximately 3,500 BDT for a handheld Bluetooth barcode scanner test unit.

17. Comprehensive Risk Analysis and Proactive Mitigation

17.1 Risk Management in the Local Retail Context

Key operational and technical risks specific to the Bangladeshi retail context are managed as follows:

ID	Technical Risk	Likelihood	Context & Mitigation Strategy
R-01	Database Concurrency Conflicts	High	Two cashiers sell the last unit of milk at the same millisecond. Mitigation: Use SQL Row-Level Locking (SELECT ... FOR UPDATE).
R-02	Internet Outages	High	Store loses connection to the central server. Mitigation: Implement a queue system on the POS that caches sales and bulk-syncs when online.
R-03	Data Entry Errors at Receiving	Medium	Clerk enters 500 instead of 50 boxes. Mitigation: System warns if GRN exceeds the Purchase Order quantity by >5%.

17.2 Comprehensive Operational & Technical Risks (R-01 to R-12)

Twelve specific operational, technical, and regulatory risks have been identified along with proactive mitigation strategies:

ID	Identified Risk Scenario	Likelihood	Impact	Proactive Mitigation Strategy
R-01	Resistance from warehouse floor workers unfamiliar with digital barcode tools	Medium	High	Design dead-simple one-touch UI; support both Bangla and English; conduct intensive 3-day on-site floor training with visual cheat sheets.

ID	Identified Risk Scenario	Likelihood	Impact	Proactive Mitigation Strategy
R-02	Inaccurate manual data entry during supplier receipt (wrong batch or expiry date)	High	High	Implement mandatory barcode scanning; enforce strict regex validation on date formats; require double-entry confirmation on bulk perishables.
R-03	Wi-Fi dead zones in metallic warehouse racking causing scan interruptions	High	Medium	Implement Service Worker caching and IndexedDB offline queue on mobile scanners; auto-sync transactions chronologically upon reconnection.
R-04	Extreme supplier lead-time unpredictability distorting safety stock calculations	High	High	Utilize Greasley's statistical model factoring both demand and lead-time variance; configure conservative service factor ($Z=1.96$ / 97.5%).
R-05	Concurrent database write collisions during simultaneous wave picking across aisles	Medium	High	Implement row-level locking (SELECT FOR UPDATE) on specific inventory batch allocations; use atomic PostgreSQL transactions to prevent double-picking.
R-06	Barcode label damage, smudging, or non-scannable supplier packages	High	Medium	Equip receiving dock with on-demand barcode label printers to re-label compromised packaging immediately upon dock inspection.

ID	Identified Risk Scenario	Likelihood	Impact	Proactive Mitigation Strategy
R-07	Inventory shrinkage masked as legitimate stock adjustment or write-off	Medium	High	Enforce dual-authorization workflow for all stock adjustments > 1,000 BDT; log all write-off reasons with mandatory photographic evidence.
R-08	System performance degradation during massive monthly cycle counting	Low	Medium	Offload heavy analytical and cycle count reconciliation queries to read-replicas or optimized materialized views refreshed asynchronously.
R-09	Scope creep across specialized third-party logistics (3PL) or automated crane robotics	High	Medium	Strictly enforce MoSCoW boundaries; classify automated robotics and external 3PL carrier integration as Phase 3 Future Scope.
R-10	Hardware terminal hardware theft or unauthorized mobile device access	Low	High	Enforce device token binding; implement automatic 15-minute idle session timeouts; restrict system access to warehouse local Wi-Fi subnet.
R-11	Data corruption or loss due to unexpected server power grid failure in Bangladesh	Medium	High	Deploy PostgreSQL with Write-Ahead Logging (WAL) and automated hourly automated snapshots backed up to cloud object storage.

ID	Identified Risk Scenario	Likelihood	Impact	Proactive Mitigation Strategy
R-12	Legal compliance breach with Bangladesh Food Safety Act 2013 on expired goods	Medium	Critical	Enforce automated system lockout that mechanically prevents generating pick-lists or dispatch notes for batches within 48 hours of expiration.

17.3 Dual-Level Verification & Validation (V&V) Testing Strategy

To satisfy the rigorous software engineering quality assurance requirements of SE-231, RetailSync enforces an exhaustive, two-tiered Verification and Validation (V&V) framework that pairs Black-Box functional evaluation with deep White-Box structural audits:

Testing Tier & Category	Target Operational Dimension	Test Case Scenario & Verification Methodology	Verifiable Pass / Acceptance Criteria
Tier 1: Black-Box Functional	Barcode Interrogation & Scanning Latency	Interrogate 1D/2D barcodes on mobile cameras and Bluetooth triggers across low and high ambient lighting (100–600 lux).	100% SKU identification accuracy; UI updates and audible/haptic feedback trigger within ≤ 350 ms.
Tier 1: Black-Box Functional	Inbound GRN vs. PO Automated Variance	Simulate delivery of 50 cartons with 3 cartons flagged as crushed; generate digital GRN against open PO.	GRN logs 47 accepted, 3 quarantined; automated supplier credit note generated; zero quarantined stock added to POS.
Tier 1: Black-Box Functional	Offline Network Resilience & Bulk Sync	Sever store ISP connection during 25 consecutive barcode checkouts; reconnect network after 5 minutes.	Local client queue caches all 25 sales with timestamps; bulk POST executes automatically on reconnect with zero dropped sales.
Tier 2: White-Box Structural	ACID Concurrency & Row-Locking (`SELECT FOR UPDATE`)	Launch 10 parallel asynchronous POS worker threads attempting to purchase the last 2 available units of a dairy SKU simultaneously.	Exactly 2 threads succeed; 8 threads receive clean out-of-stock messages; inventory balance remains exactly 0; zero database deadlocks.
Tier 2: White-Box Structural	Strict FEFO Priority Queue Verification	Seed database with 3 batches of milk expiring in 5 days, 15 days, and 45 days; issue automated checkout sales deductions.	Database triggers strictly decrement stock from the 5-day expiry batch first; automatically switches to 15-day batch only when 5-day is depleted.

Testing Tier & Category	Target Operational Dimension	Test Case Scenario & Verification Methodology	Verifiable Pass / Acceptance Criteria
Tier 2: White-Box Structural	SQL Injection & XSS Penetration Defense	Inject SQL injection strings (' OR 1=1; DROP TABLE) and XSS payloads into barcode search, user login, and PO creation fields.	All inputs sanitized via parameterized queries and ORM; malicious payloads rejected with HTTP 400; security access log records attack attempt.
Tier 2: White-Box Structural	Mathematical Replenishment Engine Precision	Execute algorithmic unit test suite across 100 historical SKU sales and lead-time distributions (EOQ, Greasley, ROP).	Computed EOQ, Safety Stock, and ROP match analytical statistical benchmarks within 0.01% floating-point tolerance.

PART F

LOOKING AHEAD

18. Scalability, Multi-Store Architecture and Product Vision

While the capstone prototype focuses on a central distribution warehouse serving 3 retail outlets, the underlying software architecture is engineered for enterprise-wide horizontal scalability:

- **Stage 1 (Capstone MVP):** Single central warehouse supporting up to 5,000 SKUs and 3 branch stores.
- **Stage 2 (Regional Expansion):** Regional distribution centers in Chittagong and Bogra with inter-warehouse stock transfer routing.
- **Stage 3 (Advanced Automation):** Integration with external automated high-density pallet shuttles and pick-to-light cart systems.
- **Stage 4 (Omnichannel Retail):** E-grocery rapid dark-store fulfillment supporting sub-30-minute hyper-local order dispatch.
- **Stage 5 (Ecosystem Integration):** Nationwide supply chain network connecting FMCG manufacturers directly to retail supermarket shelves.

18.1 Architectural Provisions for High-Volume Concurrency

To handle heavy peak transaction volumes (such as morning inbound truck unloads and afternoon store dispatch rushes), the backend services are stateless and containerized via Docker, allowing horizontal scaling behind an Nginx load balancer. PostgreSQL utilizes connection pooling (PgBouncer), strategic B-Tree indexing on SKU and batch expiry dates, row-level locking (SELECT FOR UPDATE), and read-replicas to ensure read-heavy dashboard queries never lock write-heavy floor scanning transactions.

18.2 Business Model Outline (Commercial SaaS Vision)

Post-capstone commercialization follows a tiered B2B SaaS model: 1. Starter Tier: Targeting independent medium-sized super shops with a single warehouse (monthly subscription of 15,000 BDT). 2. Enterprise Tier: Targeting national chains (Shwapno, Agora, Meena Bazar) with multi-warehouse licensing, custom ERP API connectors, and on-premise dedicated server deployment with 24/7 SLA maintenance contracts.

19. Expected Value, Financial Impact and Operational ROI

Deploying RetailSync delivers tangible, quantifiable value across four primary operational domains:

- **1. Warehouse Floor Productivity:** Eliminates paper tallying, reduces physical bin search latency by 60%, doubles picking line throughput from 40 to 95 lines/hour, and provides clear accountability.
- **2. Supply Chain & Financial ROI:** Reduces perishable food expiry wastage by an estimated 65% through enforced FEFO batch routing, saving millions of BDT annually; cuts high-velocity stockouts by 80% using dynamic EOQ/ROP mathematical triggers.
- **3. Store Operations & Customer Trust:** Ensures retail supermarket shelves remain consistently stocked with fresh goods, completely eliminating customer dissatisfaction from expired or out-of-stock items.
- **4. Academic & Pedagogical Value:** Demonstrates a complete, robust System Analysis & Design (SAD) implementation, bridging theoretical operations research formulas with modern cyber-physical web software engineering.

20. Part 1 Conclusion and Roadmap to Part 2 System Design

RetailSync moves beyond a simple CRUD application by tackling real-world operational bottlenecks: concurrency, batch expiration, and automated procurement. By grounding the project in a robust relational database architecture and strict auditability, it meets the highest standards for a System Analysis and Design capstone, offering a highly practical solution to a prominent challenge in the Bangladeshi retail sector.

This project proposal (Part 1: Project Planning and Definition) establishes the rigorous operational, algorithmic, and architectural foundation for RetailSync. Grounded in empirical supermarket logistics data from Bangladesh, the proposal formulates nine critical operational problems (P-01 to P-09), establishes six measurable S.M.A.R.T. objectives (O-01 to O-06), defines 78 detailed functional requirements (FR-01 to FR-78), and outlines mathematical inventory optimization models that bridge the gap between physical warehouse floors, POS registers, and digital enterprise decision-making.

Following academic committee evaluation and supervisor approval, the project will immediately transition into Part 2: System Architecture and Detailed Design, which will articulate Context Diagrams, Level-0 and Level-1 Data Flow Diagrams (DFDs), Entity-Relationship Diagrams (ERD), relational DDL database blueprints, and detailed UI/UX wireframes.

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PART G

APPENDICES: DETAILED SYSTEM SPECIFICATIONS

Appendix A: Detailed Functional Requirements Table (FR-01 to FR-78)

Priority Key: M = Must Have (Capstone MVP Core), S = Should Have (Important Enhancement), C = Could Have (Future/Nice to Have). The Problem Link column cross-references the permanent problem identifiers established in Section 3 (P-01 to P-09).

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-01	Account & Access	The system shall allow authorized warehouse personnel to log in securely using username/email and password with JWT token-based authentication.	M	P-09
FR-02	Account & Access	The system shall enforce Role-Based Access Control (RBAC) with predefined roles: Warehouse Operator, Receiving Clerk, Floor Supervisor, Procurement Officer, Store Manager, and System Administrator.	M	P-09
FR-03	Account & Access	The system shall support secure password reset functionality via email verification token.	M	P-09
FR-04	Account & Access	The system shall automatically terminate user sessions after 15 minutes of inactivity on mobile scanning terminals to prevent unauthorized floor access.	M	P-09

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-05	Account & Access	The system shall allow administrators to create, update, deactivate, and assign specific operational zone permissions to warehouse worker accounts.	M	P-09
FR-06	Account & Access	The system shall record every login, logout, and failed authentication attempt in an immutable security access log.	S	P-09
FR-07	Product Master	The system shall manage a centralized product catalog storing SKU code, product name, brand, barcode (EAN-13/UPC/GS1-128), unit of measurement (UOM), and packaging dimensions.	M	P-01
FR-08	Product Master	The system shall classify products into hierarchical categories: FMCG Dry Food, Perishable Produce, Chilled Dairy & Meat, Frozen Foods, Personal Care, and Household Non-Food.	M	P-02

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-09	Product Master	The system shall enforce temperature and storage classification attributes for each SKU (Ambient, Air-Conditioned 18-22°C, Chilled 2-4°C, Deep Frozen -18°C).	M	P-02
FR-10	Product Master	The system shall maintain product shelf-life parameters, including total expected shelf life in days, minimum allowable receiving shelf life (%), and retail store dispatch threshold.	M	P-03
FR-11	Product Master	The system shall allow configuration of financial costing parameters per SKU: unit purchase cost, estimated annual holding cost percentage, and fixed order processing cost.	M	P-05
FR-12	Product Master	The system shall support bulk product master data import and export via standardized CSV/Excel templates.	S	P-01
FR-13	Product Master	The system shall maintain ABC-velocity classification (Class A: High-turnover fast-moving; Class B: Medium; Class C: Slow-moving) recalculated monthly.	M	P-02

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-14	Supplier & PO	The system shall maintain a supplier directory containing company legal name, contact persons, phone numbers, trade license, lead time history, and payment terms.	M	P-07
FR-15	Supplier & PO	The system shall allow procurement officers to create, edit, and issue digital Purchase Orders (POs) linked to specific approved suppliers.	M	P-05
FR-16	Supplier & PO	The system shall automatically populate PO line items with recommended order quantities derived from the algorithmic EOQ/ROP decision engine.	M	P-05
FR-17	Supplier & PO	The system shall support a multi-tier PO approval workflow (Draft -> Pending Approval -> Approved & Issued -> Partially Received -> Completed -> Cancelled).	M	P-05
FR-18	Supplier & PO	The system shall track supplier delivery performance, automatically calculating on-time delivery rate, fulfillment accuracy, and lead-time standard deviation.	S	P-07

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-19	Supplier & PO	The system shall allow attaching supplier contracts, quality certificates, and digital invoice copies to corresponding PO records.	C	P-07
FR-20	Supplier & PO	The system shall notify suppliers and procurement officers via email when a PO is issued or approaching its agreed delivery deadline.	S	P-07
FR-21	Inbound & GRN	The system shall allow receiving clerks to look up open Purchase Orders by PO number, supplier name, or arrival schedule.	M	P-01
FR-22	Inbound & GRN	The system shall validate delivered goods against PO line items in real time via handheld barcode scanner interrogation.	M	P-01
FR-23	Inbound & GRN	The system shall record mandatory batch/lot numbers, manufacturing dates, and expiration dates for every received perishable and FMCG shipment.	M	P-03

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-24	Inbound & GRN	The system shall automatically reject or flag received batches failing to meet the minimum acceptable shelf-life threshold (e.g., less than 75% remaining life).	M	P-03
FR-25	Inbound & GRN	The system shall record damaged, spoiled, or non-conforming items with reason codes and optional photographic evidence attachment.	M	P-01
FR-26	Inbound & GRN	The system shall generate a formal digital Goods Receipt Note (GRN) upon receiving completion, categorizing accepted, rejected, and short-shipped quantities.	M	P-01
FR-27	Inbound & GRN	The system shall automatically generate credit note advisories for short-shipped or rejected quantities to reconcile supplier invoicing.	S	P-01
FR-28	Inbound & GRN	The system shall generate and print standardized warehouse internal pallet/carton barcode labels containing SKU, Batch ID, Expiry Date, and GRN number.	M	P-01

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-29	Spatial Putaway	The system shall model warehouse spatial topology using a hierarchical structure: Zone (Dry, Chilled, Bulk, Pickface) -> Aisle -> Rack -> Shelf Tier -> Bin.	M	P-02
FR-30	Spatial Putaway	The system shall assign a unique alphanumeric barcode and coordinate identifier to every physical bin location (e.g., Z1-A03-R02-S1-B04).	M	P-02
FR-31	Spatial Putaway	The system shall enforce bin volumetric and weight capacity constraints, preventing putaway suggestions into overloaded bins.	M	P-02
FR-32	Spatial Putaway	The system shall generate directed putaway instructions guiding operators to the optimal bin based on SKU velocity (Class A near dispatch docks) and temperature compatibility.	M	P-02
FR-33	Spatial Putaway	The system shall require scanning the destination bin barcode to confirm physical placement before committing the inventory location change.	M	P-02

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-34	Spatial Putaway	The system shall provide a visual 2D interactive floor map displaying real-time bin utilization, occupancy percentage, and category zoning.	S	P-02
FR-35	Spatial Putaway	The system shall allow authorized supervisors to execute internal bin-to-bin stock transfers with mandatory source and target bin scan verification.	M	P-02
FR-36	Ledger & FEFO	The system shall maintain an immutable, append-only inventory transaction ledger recording every stock movement (Inbound, Putaway, Pick, Transfer, Adjustment, Write-Off).	M	P-04
FR-37	Ledger & FEFO	The system shall compute real-time on-hand, allocated (reserved), and available-to-promise (ATP) inventory levels per SKU and batch.	M	P-04
FR-38	Ledger & FEFO	The system shall enforce strict First-Expired, First-Out (FEFO) allocation logic, automatically queuing the batch with the earliest valid expiration date for outbound picking.	M	P-03

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-39	Ledger & FEFO	The system shall support FIFO (First-In, First-Out) fallback allocation for non-perishable goods that do not carry manufacturer expiration dates.	M	P-03
FR-40	Ledger & FEFO	The system shall provide dynamic expiry alert windows, automatically classifying batches into: Safe (>60 days), Watch (31-60 days), Warning (15-30 days), and Critical (<=14 days).	M	P-03
FR-41	Ledger & FEFO	The system shall automatically lock expired or recalled batches, mechanically preventing operators from picking them and transferring them to a Quarantine Bin.	M	P-03
FR-42	Ledger & FEFO	The system shall enforce strict database ACID transactions to ensure stock decrements and allocations never result in negative stock balances.	M	P-04
FR-43	Ledger & FEFO	The system shall provide complete forward and backward batch traceability (Supplier Lot -> GRN -> Warehouse Bin -> Store Dispatch -> Retail Outlet).	M	P-09

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-44	Ledger & FEFO	The system shall support cold-chain temperature logging history linked to specific perishable inventory batches.	C	P-09
FR-45	DSS Replenishment	The system shall calculate the Economic Order Quantity (EOQ) dynamically using historical annual demand, fixed order cost, and unit holding cost.	M	P-05
FR-46	DSS Replenishment	The system shall calculate statistical Safety Stock using Greasley's dual-variance model, accounting for daily demand variance and supplier lead-time variance.	M	P-05
FR-47	DSS Replenishment	The system shall allow procurement officers to configure desired Customer Service Levels (e.g., 90%, 95%, 97.5%, 99%) mapping to statistical Z-scores.	M	P-05
FR-48	DSS Replenishment	The system shall compute Dynamic Reorder Points ($ROP = (\text{Average Daily Demand} * \text{Average Lead Time}) + \text{Safety Stock}$) for every active SKU.	M	P-05

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-49	DSS Replenishment	The system shall trigger automated replenishment alerts when an item's Inventory Position (On-Hand + On-Order - Allocated) breaches its ROP threshold.	M	P-05
FR-50	DSS Replenishment	The system shall automatically compile suggested replenishment orders into draft Purchase Orders awaiting one-click procurement officer review.	M	P-05
FR-51	DSS Replenishment	The system shall incorporate promotional demand uplift multipliers configured by merchandisers for festival periods (e.g., Eid, Ramadan, Puja).	S	P-05
FR-52	DSS Replenishment	The system shall provide simulated 'what-if' cost-benefit modeling comparing current inventory carrying costs against proposed algorithmic parameters.	C	P-05
FR-53	Outbound Picking	The system shall provide a multi-branch store portal allowing retail super shop managers to submit stock requisition orders to the central warehouse.	M	P-08

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-54	Outbound Picking	The system shall consolidate individual branch orders into optimized wave picking batches based on delivery vehicle schedules and warehouse zones.	M	P-06
FR-55	Outbound Picking	The system shall generate optimized picking sequences ordering items by aisle and rack sequence to minimize picker walking distance.	M	P-06
FR-56	Outbound Picking	The system shall require pickers to scan the bin barcode and product barcode to verify item identity and FEFO batch compliance during picking.	M	P-06
FR-57	Outbound Picking	The system shall allow operators to flag short-picked items or damaged bin stock, immediately triggering dynamic replenishment or split-order handling.	M	P-06
FR-58	Outbound Picking	The system shall support a staging and packing module where picked items are sorted into store-specific dispatch rolling cages or pallets.	M	P-06

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-59	Outbound Picking	The system shall generate an official Delivery Chalan / Dispatch Note and manifest linked to outbound transport vehicle plate numbers.	M	P-08
FR-60	Outbound Picking	The system shall allow branch store managers to digitally acknowledge receipt of outbound shipments, reconciling in-transit variances.	M	P-08
FR-61	Cycle Counting	The system shall support continuous cycle counting schedules based on ABC classification (Class A counted monthly, Class B quarterly, Class C biannually).	M	P-04
FR-62	Cycle Counting	The system shall generate blind cycle count sheets where expected system quantities are masked from counters to prevent confirmation bias.	M	P-04
FR-63	Cycle Counting	The system shall compare physical scan counts against book inventory, highlighting positive (surplus) and negative (shrinkage) discrepancies.	M	P-04

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-64	Cycle Counting	The system shall require supervisor approval and mandatory discrepancy reason codes (Breakage, Spoilage, Theft, Mislabeling) for inventory adjustments.	M	P-04
FR-65	Cycle Counting	The system shall support an unsupervised machine-learning module (Isolation Forest) to detect unusual shrinkage clusters across SKUs, shifts, and zones.	C	P-04
FR-66	Cycle Counting	The system shall generate financial shrinkage variance reports summarizing total inventory loss value over time.	S	P-04
FR-67	Reporting & BI	The system shall provide an executive dashboard rendering real-time KPIs: Total Inventory Value, GMROI, Inventory Turnover Ratio, and Stockout Frequency.	M	P-09
FR-68	Reporting & BI	The system shall render real-time floor telemetry: Active Inbound Unloading, Open Pick Waves, Bin Occupancy %, and Dispatch Dock Status.	M	P-02

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-69	Reporting & BI	The system shall generate a Perishable Expiry Watchlist report categorizing stock at risk of expiration within 7, 14, and 30 days.	M	P-03
FR-70	Reporting & BI	The system shall provide a Supplier Performance Scorecard displaying On-Time In-Full (OTIF) rates, lead-time variance, and quality rejections.	S	P-07
FR-71	Reporting & BI	The system shall provide an Outbound Store Fulfillment report analyzing branch requisition turnaround time and fulfillment accuracy.	S	P-08
FR-72	Reporting & BI	The system shall allow export of all tabular data, reports, and audit logs into PDF and Excel (.xlsx) formats.	M	P-09
FR-73	Config & Audit	The system shall maintain an immutable, tamper-evident audit log of all system configuration changes, role updates, and manual inventory overrides.	M	P-09

ID	Module Name	Detailed Functional Requirement Statement	Priority	Problem Link
FR-74	Config & Audit	The system shall record client IP address, user agent, user ID, timestamp, and before/after values for all administrative database modifications.	M	P-09
FR-75	Config & Audit	The system shall allow system administrators to configure system-wide parameters (default service level Z, expiry warning thresholds, warehouse operating hours).	M	P-09
FR-76	Config & Audit	The system shall comply with the Bangladesh Personal Data Protection Ordinance 2025, ensuring worker and supplier contact details are encrypted and access-restricted.	M	P-09
FR-77	Config & Audit	The system shall provide automated scheduled database backup utilities with off-site cloud storage synchronization.	M	P-09
FR-78	Config & Audit	The system shall support multi-lingual user interface toggling between Bangla and English for all floor operator screens.	S	P-02

Appendix B: Complete Requirements Traceability Matrix

This traceability matrix demonstrates complete, unbroken alignment between the real-world operational problems (Section 3), project objectives (Section 10), and the functional and non-functional requirements (Section 13 and Appendix A):

Problem ID	Problem Domain Description	Addressed by Objectives	Addressed by Functional Requirements	Addressed by NFRs
P-01	Disjointed Inbound Receiving & Manual GRN Verification	O-01	FR-07, FR-12, FR-21 to FR-28	NFR-04, NFR-08, NFR-09
P-02	Warehouse Opacity & Blind Spatial Putaway (Bin Latency)	O-02	FR-08, FR-09, FR-13, FR-29 to FR-35, FR-68, FR-78	NFR-04, NFR-08, NFR-11
P-03	Perishable Spoilage & Wastage from Non-FEFO Picking	O-03	FR-10, FR-23, FR-24, FR-38 to FR-41, FR-69	NFR-03, NFR-12, NFR-13
P-04	Inventory Shrinkage, Phantom Stock & Discrepancies	O-06	FR-36, FR-37, FR-42, FR-61 to FR-66	NFR-03, NFR-05, NFR-12
P-05	Inaccurate Replenishment: Over/Understocking Trap	O-04	FR-11, FR-15, FR-16, FR-17, FR-45 to FR-52	NFR-04, NFR-10, NFR-14
P-06	Outbound Picking Inefficiencies & Wave Assembly Lag	O-05	FR-54 to FR-58	NFR-04, NFR-05, NFR-08
P-07	Supplier Performance & Lead-Time Volatility Opacity	O-04	FR-14, FR-18, FR-19, FR-20, FR-70	NFR-04, NFR-10, NFR-11
P-08	Information Silos Between Central DC & Branch Stores	O-05	FR-53, FR-59, FR-60, FR-71	NFR-01, NFR-04, NFR-06
P-09	Regulatory Non-Compliance & Lack of Immutable Audit	O-06	FR-01 to FR-06, FR-43, FR-44, FR-72 to FR-77	NFR-01, NFR-02, NFR-12, NFR-13
Cross-Cutting	Enterprise Security, Concurrency & Data Privacy	O-06	FR-01 to FR-06, FR-73 to FR-77	NFR-01, NFR-02, NFR-03, NFR-05
Cross-Cutting	High Availability, Maintainability & Cost Efficiency	O-06	FR-72, FR-75, FR-77	NFR-06, NFR-07, NFR-10, NFR-14

Appendix C: Comprehensive Glossary of Terms (24 Industry Definitions)

Comprehensive domain definitions for technical, supply chain, and regulatory terminology utilized throughout RetailSync:

Domain Term / Acronym	Formal Technical & Operational Definition
SKU (Stock Keeping Unit)	A distinct type of item for sale, such as a product and all of its attributes (size, flavor, packaging). Each unique item in the super shop warehouse is assigned a unique SKU code.
GRN (Goods Receipt Note)	A formal digital or physical receipt issued by the receiving warehouse acknowledging that products delivered by a supplier have been physically received and inspected.
FEFO (First-Expired, First-Out)	An inventory allocation and picking methodology where batches with the earliest expiration dates are dispatched first, regardless of when they physically arrived in the warehouse.
FIFO (First-In, First-Out)	An inventory management method where the oldest physical stock is picked and dispatched first. Used in RetailSync as a fallback for non-perishable goods lacking manufacturer expiry dates.
EOQ (Economic Order Quantity)	A mathematical formulation that determines the ideal order quantity a company should purchase to minimize total inventory costs, balancing order setup fees against holding costs.
Safety Stock (SS)	A buffer level of extra stock held in the warehouse to mitigate the risk of stockouts caused by unpredictable fluctuations in consumer demand or supplier delivery delays.
ROP (Reorder Point)	The specific inventory level that triggers the purchase of replenishment stock. Formulated as lead time demand plus safety stock: $ROP = (d * L) + SS$.
Putaway	The physical and systematic process of moving received goods from the inbound unloading dock to their designated optimal storage bin location on the warehouse racks.
Pickface	The designated front-facing, ground-accessible warehouse rack location from which operators physically pick goods to assemble outbound store orders.
Wave Picking	A picking method where multiple store orders are grouped into consolidated 'waves' based on delivery routes or departure times to optimize picker travel efficiency.
Batch Picking	A fulfillment technique where an operator picks the total required quantity of an item across multiple orders simultaneously in a single pass, sorting them at a staging station.

Domain Term / Acronym	Formal Technical & Operational Definition
Cycle Counting	An ongoing inventory auditing procedure where a small subset of warehouse inventory is physically counted on a continuous cyclical schedule without freezing entire warehouse operations.
Shrinkage	The loss of inventory that can be attributed to factors such as employee theft, customer shoplifting, administrative paperwork errors, vendor fraud, and product breakage.
Phantom Inventory	A discrepancy where the inventory management software records stock as available on-hand, but the physical item does not actually exist in the physical bin location.
Lead Time (L)	The total elapsed latency from the moment a purchase order is submitted to a supplier until the goods are physically delivered, inspected, and made available in the warehouse.
On-Time In-Full (OTIF)	A premier supply chain KPI measuring the percentage of supplier deliveries that arrived within the agreed delivery window containing 100% of the ordered quantities.
ABC Classification	An inventory categorization method based on the Pareto Principle (80/20 rule), classifying goods into Class A (high-value / fast-moving), Class B (moderate), and Class C (low).
Quarantine Bin	A dedicated physical and logical storage area in the warehouse reserved strictly for expired, recalled, or damaged items, mechanically locked from outbound picking.
Cross-Docking	A logistics practice where inbound goods from a supplier are transferred directly to outbound store dispatch staging with minimal or zero intermediate warehouse storage.
Delivery Chalan	A legal transport document and dispatch note accompanying outbound goods delivered to retail branch stores, detailing line item quantities and transport details.
GS1-128 / EAN-13	International barcode standards used across the retail supply chain to encode product identification, batch lot numbers, weight, and expiration dates.
ACID Compliance	Database properties (Atomicity, Consistency, Isolation, Durability) ensuring reliable execution of stock transaction ledgers without concurrency collisions or negative stock balances.
PDPO (2025)	The Personal Data Protection Ordinance 2025 of Bangladesh, regulating the collection, storage, processing, and protection of personal data of warehouse staff and suppliers.
GMROI	Gross Margin Return on Investment: A retail inventory profitability evaluation metric analyzing a firm's ability to turn inventory into cash above the cost of the inventory.

